

COURSE STRUCTURE AND DETAILED SYLLABUS

R22

CIVILENGINEERING

For

B.TECH.FOUR YEAR DEGREE COURSE

(Applicable for the students admitted into E1 from the Academic Year 2022-23)

(I–IV Years Syllabus)



RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES

Basar, Nirmal, Telangana -504107

Course Structure of Credits

S.No	Course Category	Course Type	Standard Credits as per AICTE	Department Credits
1.	BSC	Basic Science Course	25	21
2.	ESC	Engineering Science Course	24	17.5
3.	HSMC	Humanities and Social Sciences including Management Course	12	14
4.	PCC	Program Core Course	48	70
5.	PEC	Program Elective Course	18	15
6.	OEC	Open Elective Course	18	9
7.	MC	Mandatory Course	0	0
8.	SIP	Seminar + Internship + Project Work + Comprehensive Viva	15	13
	Total		160	159.5

COURSE STRUCTURE (R22) FOR B.TECH.(REGULAR)

**Applicable for the students of B. Tech (Regular) from the Academic
Year 2022-23 and onwards**

FIRST YEAR (E1) – SEMESTER – I

Subject Code	Course Name	L-T-P	Credits	Category
	Linear Algebra and Calculus	3-1-0	4	BSC
	English	2-0-0	2	HSMC
	Engineering Physics	3-0-0	3	BSC
	Basic Electrical And Electronics Engineering	3-0-0	3	ESC
	Introduction To Civil Engineering	2-0-0	2	HSMC
	English Lab	0-0-2	1	HSMC
	Engineering Physics Lab	0-0-3	1.5	BSC
	Engineering Graphics	0-1-4	3	ESC
	Basic Electrical And Electronics Engineering Lab	0-0-2	1	ESC
	Induction Programme			
Total			20.5	

FIRST YEAR (E1) – SEMESTER – II

Subject Code	Course Name	L-T-P	Credits	Category
	Differential Equations and Vector Calculus	3-1-0	4	BSC
	Engineering Mechanics	4-0-0	4	ESC
	Engineering Geology	3-1-0	4	PCC
	Engineering Chemistry	3-0-0	3	BSC
	Programming For Problem Solving	3-0-0	3	ESC
	Engineering workshop	0-1-2	2	ESC
	Programming For Problem Solving Lab	0-0-2	1.5	ESC
	Engineering Chemistry Lab	0-0-3	1.5	BSC
Total		16-2-9	23	

SECOND YEAR (E2) – SEMESTER – I

Subject Code	Course Name	L-T-P	Credits	Category
	Building Materials and Concrete Technology	3-0-0	3	PCC
	Mechanics of Materials	3-1-0	4	PCC
	Fluid Mechanics	3-1-0	4	PCC
	Probability and Statistics	3-1-0	4	BSC
	Civil Engineering-Societal and Global Impact	2-0-0	2	HSMC
	Environmental Science	2-0-0	0	MC
	Mechanics of Fluids Lab	0-0-2	1	PCC
	Materials Testing Lab	0-0-2	1	PCC
	Concrete Technology Lab	0-0-2	1	PCC
	Technical Seminar-I	0-0-2	0	SIP
Total		16-3-8	20	

SECOND YEAR (E2) – SEMESTER – II

Subject Code	Course Name	L-T-P	Credits	Category
	Hydraulic Engineering	4-0-0	4	PCC
	Surveying	3-0-0	3	PCC
	Structural Analysis-I	3-0-0	3	PCC
	Design of Concrete Structures	4-0-0	4	PCC
	Water Resources Engineering-I	3-0-0	3	PCC
	Constitution of India	2-0-0	0	MC
	Hydraulic Engineering Lab	0-0-2	1	PCC
	Computer Aided Drafting of Buildings Lab	0-0-3	1	PCC
	Effective Technical Communication- I	0-0-2	1	HSMC
	Surveying Lab	0-0-2	1	PCC
	Technical Seminar-II	0-0-2	0	SIP
Total		19-0-11	21	

THIRD YEAR (E3) – SEMESTER – I

Subject Code	Course Name	L-T-P	Credits	Category
	Estimation and Costing	3-0-0	3	PCC
	Structural Analysis-II	3-0-0	3	PCC
	Highway and Traffic Engineering	3-0-0	3	PCC
	Soil Mechanics	3-0-0	3	PCC
	Design of Steel Structures	3-0-0	3	PCC
	Water Resources Engineering-II	3-0-0	3	PCC
	Highway Engineering Lab	0-0-2	1	PCC
	Soil Mechanics Lab	0-0-2	1	PCC
	Technical Seminar-III	0-0-2	0	SIP
Total		18-0-6	20	

THIRD YEAR (E3) – SEMESTER- II

Subject Code	Course Name	L-T-P	Credits	Category
	Construction Planning and Management	3-0-0	3	PCC
	Environmental Engineering	3-0-0	3	PCC
	Professional Elective-I	3-0-0	3	PEC
	Professional Elective-II	3-0-0	3	PEC
	Essence of Indian Tradition and Knowledge	2-0-0	0	MC
	Managerial Economics and Financial Analysis	3-0-0	3	HSMC
	Effective Technical Communication-II	0-0-2	1	HSMC
	Computational Analysis Lab	0-0-2	1	PCC
	Environmental Engineering Lab	0-0-2	1	PCC
	Theme Based Project		1	SIP
	Technical Seminar-IV	0-0-2	0	SIP
Total		17-0-8	19	

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FOURTH YEAR (E4) - SEMESTER- I

Subject Code	Course Name	L-T-P	Credits	Category
	Professional Law and Ethics	2-0-0	2	HSMC
	Remote Sensing and GIS	3-0-2	4	PCC
	Professional Elective-III	3-0-0	3	PEC
	Professional Elective-IV	3-0-0	3	PEC
	Professional Elective-V	3-0-0	3	PEC
	Open Elective-I	3-0-0	3	OEC
	Summer Internship	0	1	SIP
	Project-I		4	PROJ
Total		17-0-2	23	

FOURTH YEAR (E4) - SEMESTER – II

Subject Code	Course Name	L-T-P	Credits	Category
	Open Elective-II	3-0-0	3	OEC
	Open Elective-III	3-0-0	3	OEC
	Project-II	0-0-6	6	PROJ
	Comprehensive Viva		1	SIP
Total		6-0-6	13	

List of Professional Elective Courses (PECs)

S.No	Course Code	Course Title	Semester	L	T	P	Total	Credits
1	CE3211	Foundation Engineering	VI	3	0	0	3	3
2	CE3221	Railway and Airport Engineering	VI	3	0	0	3	3
3	CE3222	Finite Element Methods	VI	3	0	0	3	3
4	CE3223	Ground Water Hydrology	VI	3	0	0	3	3
5	CE4131	Solid and Hazardous Waste Management	VII	3	0	0	3	3
6	CE4132	Sustainable Construction Management	VII	3	0	0	3	3
7	CE4141	Ground Improvement Techniques	VII	3	0	0	3	3
8	CE4142	Transportation Economics	VII	3	0	0	3	3
9	CE4143	Advanced Fluid Mechanics	VII	3	0	0	3	3
10	CE4144	Rehabilitation and Retrofitting of Structures	VII	3	0	0	3	3
11	CE4251	Earthquake Engineering	VIII	3	0	0	3	3
12	CE4252	Environmental Impact Assessment	VIII	3	0	0	3	3
13	CE4253	Construction Equipment & Automation	VIII	3	0	0	3	3
14	CE4254	Pavement Construction and Management	VIII	3	0	0	3	3
15	CE4255	Pre stressed Concrete	VIII	3	0	0	3	3

Linear Algebra and Calculus

Internals: 40 Marks

L - T - P - C

Externals: 60 Marks

3 - 1 - 0 - 4

Objectives: To learn

1. Types of matrices and their properties.
2. Concept of a rank of the matrix and applying this concept to know the consistency and solving the system of linear equations.
3. Concept of Eigen values and Eigen vectors and to reduce the quadratic form to canonical form.
4. Concept of Sequence.
5. Concept of nature of series.
6. Geometrical approach to the mean values theorems and their application to the mathematical problems.
7. Evaluation of surface areas and volumes of revolutions of curves.
8. Evaluation of improper integrals using Beta and Gamma functions.
9. Partial differentiation, concept of total derivative.
10. Finding maxima and minima of function of two and three variables.

UNIT-I

Matrix Theory: Types of Matrices, Symmetric, Hermitian, Skew-Symmetry, Skew-Hermitian, Orthogonal matrices, Unitary matrices; Elementary row and column operations on a matrix, Rank of a matrix by Echelon form and Normal form, Inverse of a Non-singular matrix by Gauss-Jordan method; Consistency and solutions of system of linear equations using elementary operations, Gauss elimination method; Gauss Seidel Iteration method.

UNIT-II

Eigen values and Eigen vectors: Linear Transformation and Orthogonal Transformation; Characteristic roots and vectors of a matrix; Diagonalization of a matrix; Cayley-Hamilton theorem(without proof); finding inverse and power of a matrix by Caylay-Hamilton Theorem; Quadratic forms and Nature of the Quadratic forms; Reduction of quadratic form to canonical form by Orthogonal transformation.

UNIT-III

Sequences & Series: Definition of a sequence, limit; Convergent, Divergent and Oscillatory sequences. Convergent, Divergent and Oscillatory Series; Series of positive terms; Comparison test, p-test, D-Alembert's ratio test; Raabe's test; Cauchy's Integral test; Cauchy's root test;

Logarithmic test. Alternating series; Leibnitz test; Alternating Convergent series; Absolute and conditionally convergence.

UNIT-IV

Calculus: Mean value theorems: Roll's theorem, Lagrange's Mean value theorem, Cauchy's Mean value theorem; Taylor's and Macaurin's series with remainders, Expansions; Applications of definite integrals to evaluate surface area and volumes of revolutions of curves (Only in Cartesian coordinates): Definition of Improper Integrals and their convergence, Beta and Gamma functions and their applications.

UNIT-V

Multivariable Calculus (Partial Differentiation and applications): Definitions of Limits and continuity. Partial Differentiation; Euler's theorem; Total Derivative; Jacobian; Functional dependence and independence; Maxima and minima of functions of several variables (two and three variables) using Lagrange Multipliers.

Outcomes:

After learning the concepts of this course, the student must be able to

1. Write the matrix representation of set of linear equations and to analyze the solution of the system of equations.
2. Find the Eigen values and Eigen vectors.
3. Reduce the quadratic form to canonical form using orthogonal transformations.
4. Analyze the nature of sequence and series.
5. Solve the applications on the mean value theorems.
6. Evaluate the improper integrals using Beta and Gamma functions.
7. Find the extreme values of functions of two variables with/without constraints.

TEXTBOOKS:

1. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley and Sons, 8th Edition,
2. R.K.Jain and S.R.K.Iyengar Advanced Engineering Mathematics, Narosa Publications House.2008
3. B.S. Grewal, Higher Engineering Mathematics, Khanna Publications, 2009.

REFERENCES:

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
2. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
3. N.P. Bahl and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.

Category: **Humanities and Social Sciences including Management Courses**
 Subject Code: **18HS1101**

ENGLISH

Externals: 60 Marks

L-T-P-C

Internals: 40 Marks

2-0-0-2

Objectives:

- To equip students to study academic subjects more effectively and critically using the theoretical and practical components of English syllabus.
- Develop study skills and communication skills in formal and informal situations
- Improve the language proficiency of students in English with an emphasis on vocabulary, Grammar, Reading and Writing skills.

UNIT-I: The Raman Effect

Vocabulary Building: The Concept of Word Formation --The Use of Prefixes and Suffixes

Grammar: Identifying Common Errors in Writing with Reference to Articles and Prepositions.

Reading: Reading and Its Importance- Techniques for Effective Reading.

Basic Writing Skills: Sentence Structures -Use of Phrases and Clauses in Sentences Importance of Proper Punctuation- Techniques for writing precisely – **Paragraph writing** – Types, Structures and Features of a Paragraph - Creating Coherence-Organizing Principles of Paragraphs in Documents.

UNIT-II: Ancient Architecture in India

Vocabulary: Synonyms and Antonyms

Grammar: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement.

Reading: Improving Comprehension Skills – Techniques for Good Comprehension

Writing: Format of a Formal Letter-**Writing Formal Letters** E.g., Letter of Complaint, Letter of Requisition, Job Application with Resume

UNIT-III: Blue Jeans

Vocabulary: Acquaintance with Prefixes and Suffixes from Foreign Languages in English to form Derivatives-Words from Foreign Languages and their Use in English.

Grammar: Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses.

Reading: Sub-skills of Reading- Skimming and Scanning

Writing: Nature and Style of Sensible Writing- **Defining- Describing** Objects, Places and Events – **Classifying-** Providing Examples or Evidence

UNIT-IV: What Should You Be Eating

Vocabulary: Standard Abbreviations in English

Grammar: Redundancies and Clichés in Oral and Written Communication.

Reading: Comprehension- Intensive Reading and Extensive Reading

Writing: Writing Practices--Writing Introduction and Conclusion - Essay Writing-Précis Writing.

UNIT-V: How a Chinese Billionaire Built Her Fortune

Vocabulary: Technical Vocabulary and their usage

Grammar: Common Errors in English

Reading: Reading Comprehension-Exercises for Practice

Writing: Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports - Writing a Report.

Outcomes:

Upon successful completion of this course, the student shall be able to

- Use English Language effectively in spoken and written forms.
- Comprehend the given texts and respond appropriately.
- Communicate confidently in various contexts and different cultures.
- Acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.

Text Book:

1. Sudarshana, N.P. and Savitha, C. (2018). English for Engineers. Cambridge University Press.

References:

1. Swan, M. (2016). Practical English Usage. Oxford University Press.
2. Kumar, S and Lata, P (2018). Communication Skills. Oxford University Press.
3. Wood, F.T. (2007) Remedial English Grammar. Macmillan.
4. Zinsser, William. (2001). On Writing Well. Harper Resource Book.
5. Hamp-Lyons, L. (2006) Study Writing. Cambridge University Press.
6. Exercises in Spoken English. Parts I –III. CIEFL, Hyderabad. Oxford University Press

ENGINEERING PHYSICS

Internals: 40 Marks

L - T - P - C

Externals: 60 Marks

4 - 0 - 0 - 4

UNIT I:

Motion of a Particle: Basic concepts of scalars and vectors. Vector resolution. Transformation of scalars and vectors under rotation transformation. Extension to cylindrical and spherical coordinates. Newton's law and its completeness in describing particle motion. Some applications of Newton's law.

UNIT II:

Introduction to Engineering Mechanics, Force Systems Basic concepts, Particle equilibrium in 2-D & 3-D; Rigid Body equilibrium; System of Forces, Coplanar Concurrent Forces, Components in Space – Resultant- Moment of Forces and its Application; Couples and Resultant of Force System, Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Systems and Spatial Systems; Static Indeterminacy.

UNIT III:

Friction, Types of friction, Limiting friction, Laws of Friction, Static and Dynamic Friction; Motion of Bodies, wedge friction;

UNIT -IV

Centroid and Moment of Inertia of Areas: Centroids of lines, areas and volumes, Areas and volumes of revolution, Pappu's theorems and their applications, Area moment of inertia, Product moment of Inertia, Composite areas, radius of gyration.

Centre of Gravity and Mass moment of inertia: Centre of Gravity and mass moment of inertia for solids and composite bodies. Radius of gyration.

UNIT V:

Mechanical Vibrations: Basic terminology, free and forced vibrations, resonance and its effects; Degree of freedom; Derivation for frequency and amplitude of free vibrations without damping and single degree of freedom system, simple problems, types of pendulum, use of simple, compound and torsion pendulums

Reference books:

(i) Engineering Mechanics, 2nd ed. – MK Harbola

(ii) Introduction to Mechanics – MK Verma

- (iii) An Introduction to Mechanics – D Kleppner & R Kolenkow
- (iv) Principles of Mechanics – JL Synge & BA Griffiths
- (v) Mechanics – JP Den Hartog
- (vi) Engineering Mechanics – Dynamics, 7th ed. - JL Meriam
- (vii) Mechanical Vibrations – JP Den Hartog
- (viii) Theory of Vibrations with Applications – WT Thomson

Category: **Engineering Science Course**

Subject Code: **19EE1102**

BASIC ELECTRICAL AND ELETRONICS ENGINEERING

(For Chemical and Civil)

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-1-0-4

Objectives:

This course introduces the concept of

1. Electrical DC and AC circuits, basic laws of electricity and methods to solve the electrical networks
2. Construction operational features of energy conversion devices i.e. transformers, DC motors and induction motors.
3. Basics of electronics, semiconductor devices and their characteristics and operational features.

UNIT- I: (10 hours)

Dc Circuit Analysis: Electrical circuit elements- R-L-C Parameters, V–I relationship for Passive elements, Diode, Voltage and Current Independent and Dependent Sources

Circuit analysis: Kirchoff's Laws, Network reduction techniques – series, parallel, series parallel, star-to-delta, delta-to-star transformation, Source Transformation, Mesh Analysis and Nodal Analysis

Network Theorems - Thevenin's, Norton's, Maximum Power Transfer, Superposition

Step response of RL, RC and RLC circuits

UNIT- II: (10 hours)

Ac Circuit Analysis: Single Phase AC Circuits - R.M.S. and Average values, Form Factor, steady state analysis of series, Parallel and Series parallel Combinations of R, L and C with Sinusoidal excitation, concept of reactance, Impedance, Susceptance and Admittance – phase and phase difference, Concept of Power Factor, j-notation, complex and Polar forms of representation.

Resonance – Series resonance and Parallel resonance circuits

UNIT- III:(6 Hrs)

Three Phase Ac Circuits: Three phase EMF generation, delta and Y connections, line and phase quantities, solution of three phase circuits, balanced supply voltage and balanced load, phasor diagram, measurement of power in three phase circuits

UNIT- IV: (10 hours)

Basic Electronics: Introduction to electronics and electronic systems, Diode and Rectifier circuits (Half and Full wave), BJT, Transistor biasing. Small signal transistor amplifiers (CE), Operational amplifiers and their basic application, Introduction to digital circuits

UNIT- V: (12 Hrs)

Electrical Machines: Transformers-Construction, EMF equation, ratings, phasor diagram on no load and full load, equivalent circuit, regulation and efficiency calculations, open and short circuit test, applications

DC machines: Construction, EMF and Torque equations, Characteristics of DC generators and motors, applications

Induction motors: The revolving magnetic field, principle of operation, ratings, equivalent circuit, Torque-speed characteristics, applications

Outcomes: At the end of the course the student will be able to:

1. Understand the basic concept of electrical circuits under DC and AC excitation and solve basic electrical circuit problems
2. Understand basic concept and performance of transformers and motors used as various industrial drives

TEXT BOOKS:

1. Electrical Technology- Hughes Prentice Hall, 7th edition
2. Problems in Electrical Engineering- S. Parker Smith, 9 edition
3. Electronic Devices and Circuits – R.L. Boylestad and Louis Nashelsky, PEI/PHI, 9th Ed, 2006.
4. Millman's Electronic Devices and Circuits – J.Millman and C.C.Halkias, Satyabratajit, TMH, 2/e, 1998.

5. Engineering circuit analysis- by William Hayt and Jack E. Kemmerly, Mc Graw Hill Company, 6th edition.
6. Electric Machines –by I.J.Nagrath & D. P Kothari, Tata Mc Graw Hill, 7th Edition. 2005

REFERENCES:

1. Electronic Devices and Circuits - K. Lal Kishore, B.S. Publications, 2nd Edition, 2005.
2. Electronic Devices and Circuits – Anil K. Maini, Varsha Agarwal –Wiley India Pvt. Ltd. 1/e 2009.
4. Network Theory by N.C.Jagan & C.Lakshminarayana, B.S. Publications.
5. Network Theory by Sudhakar, Shyam Mohan Palli, TMH.
6. Electrical machines-PS Bhimbra, Khanna Publishers.

Category : **HSMC**

Subject Code: **19CE11**

INTRODUCTION TO CIVIL ENGINEERING

Externals: 60Marks

L-T-P-C

Internals: 40Marks

2-0-0-2

Objectives:

- To give an understanding to the students of the vast breadth and numerous areas of engagement available in the overall field of Civil Engineering
- To motivate the student to pursue a career in one of the many areas of Civil Engineering with deep interest and keenness.
- To expose the students to the various avenues available for doing creative and innovative work in this field by showcasing the many monuments and inspiring projects of public utility.

UNIT-I: (4 hours)

Introduction to civil engineering: Basics of Civil Engineering; Broad disciplines of Civil Engineering; Importance of Civil Engineering, Possible scopes for a career

History of Civil engineering: Early constructions and developments over time; Ancient monuments & Modern marvels; Development of various materials of construction and methods of construction; Works of Eminent civil engineers

Overview of National Planning for Construction and Infrastructure Development; Position of construction industry vis-à-vis other industries, five year plan outlays for construction; current budgets for infrastructure work

UNIT-II: (5 hours)

Fundamentals of Architecture & Town Planning: Aesthetics in Civil Engineering, Examples of great architecture, fundamentals of architectural design & town planning; Building Systems (HVAC, Acoustics, Lighting, etc.); LEED ratings; Development of Smart cities

Fundamentals of Building Materials: Stones, bricks, mortars, Plain, Reinforced & Prestressed Concrete, Construction Chemicals; Structural Steel, High Tensile Steel, Carbon Composites; Plastics in Construction; 3D printing; Recycling of Construction & Demolition wastes

Surveying & Geomatics: Traditional surveying techniques, Total Stations, Development of Digital Terrain Models; GPS, LIDAR

Traffic & Transportation Engineering: Investments in transport infrastructure development in India for different modes of transport; Developments and challenges in integrated transport development in India: road, rail, port and harbour and airport sector; PPP in transport sector; Intelligent Transport Systems; Urban Public and Freight Transportation; Road Safety under heterogeneous traffic; Sustainable and resilient pavement materials, design, construction and management; Case studies and examples

UNIT-III: (8 hours)

Environmental Engineering & Sustainability: Water treatment systems; Effluent treatment systems; Solid waste management; Sustainability in Construction

Geotechnical Engineering: Basics of soil mechanics, rock mechanics and geology; various types of foundations; basics of rock mechanics & tunneling

Structural Engineering: Types of buildings; tall structures; various types of

bridges; Water retaining structures; Other structural systems; Experimental Stress Analysis; Wind tunnel studies

Repairs & Rehabilitation of Structures: Basics of corrosion phenomena and other structural distress mechanisms; some simple systems of rehabilitation of structures; Non- Destructive testing systems; Use of carbon fiber wrapping and carbon composites in repairs

UNIT-IV: (6 hours)

Hydraulics, Hydrology & Water Resources Engineering: Fundamentals of fluid flow, basics of water supply systems; Underground Structures; Underground Structures Multi- purpose reservoir projects

Ocean Engineering: Basics of Wave and Current Systems; Sediment transport systems; Ports & Harbors and other marine structures

Power Plant Structures: Chimneys, Natural & Induced Draught Cooling towers, coal handling systems, ash handling systems; nuclear containment structures; hydro power projects

UNIT-V: (8 hours)

Basics of Construction Management & Contracts Management: Temporary Structures in Construction; Construction Methods for various types of Structures; Major Construction equipment; Automation & Robotics in Construction; Modern Project management Systems; Advent of Lean Construction; Importance of Contracts Management

Computational Methods, IT, IOT in Civil Engineering: Typical software used in Civil Engineering- Finite Element Method, Computational Fluid Dynamics; Computational Geotechnical Methods; highway design (MX), Building Information Modeling; Highlighting typical available software systems

Industrial lectures: Case studies of large civil engineering projects by industry professionals

Outcomes:

- Identifying the various areas available to pursue and specialize within the overall field of Civil Engineering
- Highlighting the depth of engagement possible within each of these areas

- Understanding the vast interfaces this field has with the society at large
- Highlighting possibilities for taking up entrepreneurial activities in this field
- Providing a foundation for the student to launch off upon an inspired academic pursuit into this branch of engineering

Category: **Humanities and Social Sciences including Management Courses**

Subject Code: **18HS1701**

ENGLISH LAB

Orals (Written): 50Marks

L-T-P-C

Written (Externals): 50Marks

0-0-2-1

Objectives:

- To facilitate computer-assisted multi-media instruction enabling individualized and independent language learning.
- To sensitize students to the nuances of English speech sounds, word accent, intonation and rhythm.
- To bring about a consistent accent and intelligibility in students' pronunciation of English by providing an opportunity for practice in speaking
- To improve the fluency of students in spoken English and neutralize their mother tongue influence
- To train students to use language appropriately for public speaking and interviews

Listening Skills:

Objectives:

1. To enable students develop their listening skills so that they may appreciate its role in the LSRW skills approach to language and improve their pronunciation

2. To equip students with necessary training in listening so that they can comprehend the speech of people of different backgrounds and regions

Students should be given practice in listening to the sounds of the language, to be able to recognize them and find the distinction between different sounds, to be able to mark stress and recognize and use the right intonation in sentences.

- Listening for general content
- Listening to fill up information
- Intensive listening
- Listening for specific information

Speaking Skills:

Objectives:

1. To involve students in speaking activities in various contexts
 2. To enable students express themselves fluently and appropriately in social and professional contexts
- Oral practice: Just A Minute (JAM) Sessions
 - Describing objects/situations/people
 - Role play – Individual/Group activities

UNIT – I

Understand: Listening Skill- Its importance – Purpose- Process- Types- Barriers of Listening - Communication at Work Place- Spoken vs. Written language.

Practice: Introduction to Phonetics – Speech Sounds – Vowels and Consonants -Ice-Breaking Activity and JAM Session- Situational Dialogues – Greetings – Taking Leave – Introducing Oneself and Others.

UNIT – II

Understand: Structure of Syllables – Word Stress and Rhythm– Weak Forms and Strong Forms in Context- Features of Good Conversation – Non-verbal Communication.

Practice: Basic Rules of Word Accent - Stress Shift - Weak Forms and Strong Forms in Context-Situational Dialogues – Role-Play- Expressions in Various Situations –Making Requests and Seeking Permissions - Telephone Etiquette.

UNIT – III

Understand: Intonation-Errors in Pronunciation-the Influence of Mother Tongue (MTI)-How to make Formal Presentations.

Practice: Common Indian Variants in Pronunciation – Differences in British and American Pronunciation- Formal Presentations.

UNIT – IV

Understand: Listening for General Details-Public Speaking – Exposure to Structured Talks.

Practice: Listening Comprehension Tests- Making a Short Speech – Extempore

UNIT – V

Understand: Listening for Specific Details- Interview Skills.

Practice: Listening Comprehension Tests- Mock Interviews.

Outcomes: Students will be able to attain

- Better understanding of nuances of English language through audio- visual experience and group activities
- Neutralization of accent for intelligibility

- Speaking skills with clarity and confidence which in turn enhances their employability skills

Category: **Basic Science Course**

Subject Code: **19PH1701**

ENGINEERING PHYSICS LAB

Internals: 40 Marks

L - T - P - C

Externals: 60 Marks

0 - 0 - 3 - 1.5

List of Experiments:

1. Fly Wheel Experiment
2. Torsional Pendulum experiment
3. Melde's Experiment Setup
4. Principle of Moments
5. Coefficient of Friction- Inclined Plane
6. Normal and Tension Forces
7. Concurrent Forces
8. Spring Damper System – Simulation

Category: **Engineering Science Course**

Subject Code: **19CE1701/1801**

ENGINEERING GRAPHICS

(Common to all Branches except ME)

Externals: 60Marks

L-T-P-C

Internals: 40Marks

2-0-4-3

Course Objectives:

- To introduce the students to the “Universal Language of Engineers” for effective communication through drawing.
- To understand the basic concepts of drawing through modern techniques.
- To impart knowledge about standard principles of projection of objects.
- To provide the visual aspects of Engineering drawing using Auto-CAD.

UNIT-I:(6 Hours)

Civil Engineering

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Introduction to Engineering Drawing: Principles of Engineering Graphics and their significance, usage of Drawing instruments, lettering, types of lines and Dimensioning.

Geometrical constructions: Construction of regular polygons.

Introduction to scales, Conic sections and Cycloidal curves

(Conventional)

UNIT-II: (8 Hours)

Orthographic projections: Principles of Orthographic Projections

Projections of Points: Projections of Points placed in different quadrants

Projection of lines: lines parallel and inclined to both the planes (Determination of true lengths and true inclinations and traces) (Conventional)

UNIT-III: (6 Hours)

Fundamentals of AutoCAD: Introduction to Auto-CAD, DRAW tools, MODIFY tools, TEXT, DIMENSION, PROPERTIES

Projection of planes: Planes inclined to both the reference planes
(Auto-CAD)

UNIT-IV: (9 Hours)

Projection of Solids: Projection of solids whose axis is parallel to one of the reference planes and inclined to the other plane, axis inclined to both the planes (Auto-CAD)

Projection of sectioned solids: Sectioning of simple solids like prism, pyramid, cylinder and cone in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other – obtaining true shape of section.

(Auto-CAD)

UNIT-V: (9 Hours)

Development of surfaces: Development of surfaces of Right Regular Solids - Prism, Pyramid, Cylinder and Cone (Auto-CAD)

Isometric Projections: Principles of Isometric projection – Isometric Scale, Isometric Views of planes and simple solids (Auto-CAD)

Perspective projections: Basic concepts of perspective views.

Course Outcomes:

At the end of the course, the student will be able to

- Use Engineering principles and techniques to understand and interpret engineering drawings.
- Understand the concepts of Auto-CAD.
- Draw orthographic projections of lines, planes and solids using Auto-CAD.

- Use the techniques, skills and modern engineering tools necessary for engineering practices.

Text/Reference Books:

1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House
2. Gopalakrishna K.R., “Engineering Drawing” (Vol. I&II combined), Subhas Stores, Bangalore, 2007.
3. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education
4. Venugopal K. and Prabhu Raja V., “Engineering Graphics”, New Age publications
5. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication
6. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers
7. (Corresponding set of) CAD Software Theory and User Manuals

SCHEME OF EVALUATION		
Assessment Method	Description	Weightage
Assignment & Continuous monitoring	Assignments & Evaluation of drawings	25%
Mid Term	2 mid examinations will be conducted and best of 2 will be considered.	25%
End Term	Students will be evaluated based on the understanding of the principles, skills and practices of the course	50%

Category: **Engineering Science Course**

Subject Code: **19EE1702**

BASIC ELECTRICAL & ELECTRONICS ENGINEERING LABORATORY

Externals: 60Marks

L-T-P-C

Civil Engineering

Page **23** of **125**

Objectives:

- To expose the students to the concepts of electrical and electronics circuits and their applications
- To expose the students to the operation of dc machines and transformer and give them experimental skills.

List of Experiments:

List of Laboratory Experiments/Demonstrations (any eight of the following):

1. Introduction to Lab:

(a) Basic safety precautions. Introduction and use of measuring instruments – voltmeter, ammeter, multi-meter, oscilloscope. Real-life resistors, capacitors and inductors.

(b) *Demonstration of cut-out sections of machines: dc machine (commutator-brush arrangement), induction machine (squirrel cage rotor), synchronous machine (field winding - slip ring arrangement) and single-phase induction machine.*

2. Measuring the steady-state and transient time-response of R-L, R-C, and R-L-C circuits to a step change in voltage (transient may be observed on a storage oscilloscope). Sinusoidal steady state response of R-L, and R-C circuits – impedance calculation and verification. Observation of phase differences between current and voltage. Resonance in R-L-C circuits.
3. Transformers: Observation of the no-load current waveform on an oscilloscope (non sinusoidal wave-shape due to B-H curve nonlinearity should be shown along with a discussion about harmonics). Loading of a transformer: measurement of primary and secondary voltages and currents, and power.
4. Torque Speed Characteristic of separately excited dc motor.
5. Torque-Slip Characteristic of an induction motor.
6. Characteristic of the lamps (Tungsten, Fluorescent and Compact Fluorescent Lamps)
7. Verification of Network Theorems
8. V-I characteristics of Diodes and BJT
9. Half-wave and full-wave rectifiers, rectification with capacitive filters, zener diode
10. Studies on logic gates
11. Small signal transistor amplifiers (CE)

Outcome:

Upon successful completion of this lab, the student will be able to

- Understand principles of measuring instruments of voltage, current and power
- Analyse the characteristics of semiconductor devices and understand their applications
- Analyse the characteristics and evaluate performance of DC machines and transformers

Induction Programme

Category: **Basic Science Course**

Subject Code: **18MA1201**

Differential Equations and Vector Calculus

Internals: 40 Marks

L - T - P - C

Externals: 60 Marks

3 - 1 - 0 - 4

Objectives:

- Methods of solving the differential equations of first and higher order.
- To study the methods of solving improper integrals and the concepts of multiple integrals
- The basic properties of vector valued functions and their applications to line, surface and volume integrals
- To study numerical methods to analyze an experimental data.

UNIT-I

Ordinary Differential Equations of first order: Exact first order differential equation, finding integrating factors, linear differential equations, Bernoulli's, Riccati, Clairaut's differential equations, finding orthogonal trajectory of family of curves, Newton's Law of Cooling, Law of Natural growth or decay.

UNIT-II

Ordinary Differential Equations of higher order: Linear dependence and independence of functions, Wronskian of n- functions to determine Linear Independence and dependence of

functions, Solutions of Second and higher order differential equations (homogeneous & non-homogeneous) with constant coefficients, Method of variation of parameters, Euler-Cauchy equation.

UNIT-III

Integral Calculus: Convergence of improper integrals, tests of convergence, Beta and Gamma functions - elementary properties, differentiation under integral sign, differentiation of integrals with variable limits - Leibnitz rule. Rectification, double and triple integrals, computations of surface and volumes, change of variables in double integrals - Jacobians of transformations, integrals dependent on parameters – applications.

UNIT-IV

Vector Differentiation: Vector point functions and scalar point functions. Gradient, Divergence and Curl. Directional derivatives, Tangent plane and normal line. Vector Identities. Scalar potential functions. Solenoidal and Irrotational vectors.

UNIT-V

Vector Integration: Line, Surface and Volume Integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications. Numerical Methods: Introduction and motivation about numerical methods, True value, approximate value, error, error percentage, algebraic equations, transcendental equations, Newton-Raphson method, Bisection method.

Outcomes: At the end of the course student will be able to

- Solve first order linear differential equations and special nonlinear first order equations like Bernoulli, Riccati & Clairaut's equations
- Compute double integrals over rectangles and type I and II" regions in the plane
- Explain the concept of a vector field and make sketches of simple vector fields in the plane.
- Explain concept of a conservative vector field, state and apply theorems that give necessary and sufficient conditions for when a vector field is conservative, and describe applications to physics.
- Recognize the statements of Stokes' Theorem and the Divergence Theorem and understand how they are generalizations of the Fundamental Theorem of Calculus.
- Solve the problems in diverse fields in engineering science using numerical methods.

Suggested Reading:

1. Advanced Engineering Mathematics (3rd Edition) by R. K. Jain and S. R. K. Iyengar, Narosa Publishing House, New Delhi
2. 1. Advanced Engineering Mathematics (8th Edition) by Erwin Kreyszig, Wiley-India.
3. Dr. M.D. Raisinghania, Ordinary and Partial differential equations S.CHAND, 17th edition

4.

Category: Engineering Science Course

Subject Code: 19CE1201

ENGINEERING MECHANICS

Internals: 40 Marks

Externals: 60 Marks

L – T – P – C

4 – 0 – 0 – 4

Course Objectives:

- To understand force, moment equilibrium and free body diagrams of various structural systems.
- Apply fundamental concepts of statics, kinematics and kinetics of particle to analyze simple, practical problems.
- To know basic concepts of loads, their behavior, analysis of bodies.
- To determine the work energy principles and impulsive momentum theory.

UNIT - I

Introduction to Engineering Mechanics - Force Systems: Basic concepts, Particle equilibrium in 2-D & 3-D; Rigid Body equilibrium; System of Forces, Coplanar Concurrent Forces, Components in Space – Resultant- Moment of Forces and its Application; Couples and Resultant of Force System, Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Systems and Spatial Systems; Static Indeterminacy.

UNIT - II

Friction: Types of friction, Limiting friction, Laws of Friction, Static and Dynamic Friction; Motion of Bodies, ladder friction Centroid and Centre of Gravity -Centroid of Lines, Areas and Volumes from first principle, centroid of composite sections; Centre of Gravity.

UNIT - III

Area moment of inertia- Definition, Moment of inertia of plane sections from first principles, Theorems of moment of inertia, Moment of inertia of standard sections and composite sections; Product of Inertia, Parallel Axis Theorem, Perpendicular Axis Theorem. Mass Moment of Inertia: Moment of Inertia of Masses - Transfer Formula for Mass Moments of Inertia – Mass moment of inertia of composite bodies.

UNIT - IV

Kinematics of Particles: Kinematics of particles – Rectilinear motion – Curvilinear motion – Projectiles. Kinetics of Particles: Kinetics of particles – Newton's Second Law – Differential equations of rectilinear and curvilinear motion – Dynamic equilibrium – Inertia force – D. Alembert's Principle applied for rectilinear and curvilinear motion.

UNIT - V

Work - Energy Principle: Equation of translation, principle of conservation of energy, work - energy principle applied to particle motion and connected systems, fixed axis rotation. Impulse – Momentum Principle: Introduction, linear impulse momentum, principle of conservation of linear momentum, elastic impact and types of impact, loss of kinetic energy, coefficient of restitution.

TEXT BOOKS:

1. Shames and Rao (2006), Engineering Mechanics, Pearson Education
2. Reddy Vijay Kumar K. and J. Suresh Kumar (2010)

Engineering Mechanics – Statics & Dynamics REFERENCE BOOKS:

1. Timoshenko S.P and Young D.H., "Engineering Mechanics", McGraw Hill International Edition, 1983.
2. Andrew Pytel, Jaan Kiusalaas, "Engineering Mechanics", Cengage Learning, 2014.
3. Beer F.P & Johnston E.R Jr. Vector, "Mechanics for Engineers", TMH, 2004.

4. Hibbeler R. C & Ashok Gupta, “Engineering Mechanics”, Pearson Education, 2010.
5. Tayal A.K., “Engineering Mechanics – Statics & Dynamics”, Umesh Publications, 2011.
6. Basudeb Bhattacharyya, “Engineering Mechanics”, Oxford University Press, 2008.
7. Meriam. J. L., “Engineering Mechanics”, Volume-II Dynamics, John Wiley & Sons, 2008.
8. P.C Dumir et al. “Engineering Mechanics”, University press

Category: Professional Core Course

Subject Code:19CE2202

ENGINEERING GEOLOGY

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-1-0-4

Objectives:

- To understand all spheres of the Earth.
- To understand the formation of different types of rocks and rock-cycle.
- To assess the geological structures like faults, folds, joints etc.
- To study the concept of weathering, process of formation of soil
- To understand different Geological hazards like Earthquakes, Volcanoes, and Landslides.
- To conceptualize the site investigation methods to know the ground conditions for dam sites, tunnels and other structures.

Unit-I: (12 hours)

Introduction: Geology in Civil Engineering, Branches of Geology, Solid Earth: Shape, size, interior of the solid earth. Spheres of Earth (Lithosphere, Hydrosphere, Cryosphere and Atmosphere) and their interactions; Earth Processes and their interactions: Plate Tectonics, Volcanism.

Mineralogy: Definition, silicate and non silicate minerals, crystalline structure, physical properties in mineral identification, rock forming minerals and their identification.

Unit-II (12 hours)

Petrology: Formation and classification of rocks, Igneous rocks: mineral composition, textures, reasons of textural variation, classification of igneous rocks, Sedimentary rocks: rock weathering, decomposition and disintegration, textures of sedimentary rocks and classification. Metamorphic rocks: agents and types of metamorphism, metamorphic textures and structures their types,

physical properties in rock identification, rock cycle. Engineering properties of rocks: Drilling, Core recovery, tests on rock samples - compression, tensile, shear and slake durability tests.

Structural Geology: Outcrop, Strike and dip, folds and types, faults and their types and classification, Geomorphology and their features of river water, ocean water.

UNIT-III (8 hours)

Geology of soils: Soil, Formation, Engineering and Genetic Classification of soils, types of Indian soils, description and engineering use of soils, soil development, clay mineralogy.

Hydro Geology: Hydrologic cycle, rock and water interaction, Water table, aquifers, occurrence of ground water in various lithological formations, ground water movement, springs, ground water exploration – surface and sub surface investigations, ground water provinces of India.

UNIT-IV (10 hours)

Geology of Dams and Reservoirs: Types of Dams, Engineering and Geological investigations for a dam site, Analysis of dam failure; Ground improvement techniques, Reservoir – site selection, leakages, detection, Engineering Geology of major Dam sites of India.

UNIT-V (10 hours)

Tunnels: Purpose of tunneling, problems in tunneling, Lining, Engineering and Geological investigations of tunnels in rock, Stand up time of different rocks, pay line and over break, logging of tunnels and geology of some well-known Indian tunnels.

Geological Hazards: Disaster management - prevention, mitigation and resource planning, Geological aspects of Earthquakes, tsunamis and Landslides

Outcomes:

- Global vision of spheres of Earth and their interactions.
- Understood the formation of different rocks and rock cycle in detail.
- Can identify the minerals and rocks based on their physical properties.
- Can assess and analyze the geological features like folds, faults etc.
- Understood the concept of Weathering and formation of soils.
- Knows the reasons for several geological disasters like Earthquakes, Landslides, Subsidence, Plate Tectonics and Volcanism.
- Understood the considerations for site selection for engineering projects.
- Knows the site investigation methods and the ground conditions to be considered for dam sites, tunnels and other structures
- Will gain knowledge about the groundwater availability zones and groundwater management.

References:

1. K.V.G.K. Gokhale, "Principles of Engineering Geology", BS Publications, Hyderabad, 2005.
2. David George Price, "Engineering Geology: Principles and Practice", Springer, 2009.
3. Chennakesavulu, N., "Text book of Engineering Geology", Mac Millan Ltd., New Delhi, 2009.
4. Parbin Singh., "Engineering and General Geology", Katson Publishers, 2009

Category: **Basic Science Course**

Subject Code: **19CY1204**

ENGINEERING CHEMISTRY**Objectives:**

- To bring adaptability to the concepts of chemistry and to acquire the required skills to become a perfect engineer.
- To impart the basic knowledge of kinetic modifications which make the student to understand the technology based on them.
- To acquire the knowledge of electrochemistry, corrosion and water treatment which are essential for the Engineers and in industry.
- To acquire the skills pertaining to spectroscopy and to apply them for medical field etc.
- To impart then knowledge of engineering materials especially focusing more into nanomaterials.

UNIT 1: (7 hours)

Spectroscopy: Introduction to spectroscopy, electromagnetic radiations, different types of spectroscopy, principle of spectroscopy, spectrophotometer Microwave spectroscopy: principle, microwave spectra of diatomic molecules, selection rules for microwave spectra, applications of microwave spectroscopy: determination of bond length, dipole moment measurement, determination of isotopic mass of an element. Infrared spectroscopy: introduction and principles of IR, types of vibrations: bending and stretching, Hooke's law for stretching vibrations, characteristic frequencies of common functional groups, IR instrumentation, interpretation and applications of IR spectrum with examples.

Ultra-violet spectroscopy: Introduction and principle of UV spectroscopy, color interpretation with VBT and MOT, types of electronic transitions, selection rules, chromophores and auxochromes

with examples, conjugation effect, absorption and intensity shifts, applications of UV spectroscopy.

UNIT II: (14 hours)

Chemical kinetics: Complex reactions: definition and classification of complex reactions, definition of reversible reactions with examples, rate law derivation for reversible reactions.

Consecutive reactions: definition, rate law derivation and examples of consecutive reactions.

Parallel reactions: definition, rate law derivation and examples of parallel reactions.

Steady-state approximation: introduction, kinetic rate law derivation by applying steady state approximation in case of the oxidation of NO and pyrolysis of methane.

Chain reactions: introduction, types and mechanism of chain reactions, stationary and non-stationary chain reactions with examples, deriving the kinetic rate equation using a general chain reaction.

Photochemical reactions: introduction, Stark-Einstein law of photochemical equivalence, photophysical processes: IC, ISC, fluorescence and phosphorescence with examples, kinetic rate law derivation in case of photochemical decomposition of HI and photochemical combination of H_2 and Br_2 .

Electrochemistry: Types of electrodes: introduction, metal-metal ion electrodes, metal-insoluble salt-anion electrodes, calomel electrode, gas-ion electrodes, hydrogen and chlorine electrodes, oxidation-reduction electrodes, amalgam electrodes.

Types of cells: classification into chemical and concentration cells, chemical cells with transference and without transference, classification of concentration cells into electrolyte and electrode concentration cells, electrolyte concentration cells with and without transference, amalgam and gas concentration cells, examples for these cells.

EMF and applications of EMF: determination of pH, determination of the valency of the ions, potentiometric titrations. pH: definition of pH and determination of pH by various methods, acid-base titrations.

Thermodynamic data: enthalpy and entropy of cell reactions, Gibbs-Helmholtz equation and applications.

UNIT IV: (14 classes)

Corrosion and its prevention: Mechanism of Dry and wet corrosion (rusting of iron),

Types of corrosion, galvanic corrosion, stress corrosion, pitting and crevice corrosion.

Factors affecting corrosion,

preventive measures(proper design, Cathodic and Anodic protection, Electroplating, tinning, galvanizing.

Water Chemistry: Water chemistry, Introduction to Hardness and alkalinity of water, degree of hardness, units of hardness, numerical problems, associated with it. Alkalinity: Causes, Determination and calculation of alkalinity, numerical problems associated. Disinfection, boiler feed water, priming and foaming and its treatment and related.

UNIT V:(9 classes)

Engineering Materials: Polymers-Types of Polymerization (Chain & Step growth) Plastics: Thermoplastic & Thermo setting resins; Preparation, properties, engineering applications of PVC, Teflon and Bakelite.

Lubricants: Classification with examples-Characteristicsof a good lubricant & mechanism of lubrication (thick film, thin film and extreme pressure) –properties of lubricants: viscosity , Cloud point, flash and fire points.

Refractories: Classification, characteristics of a good refractory and applications.

Cement: Classification, composition, preparation, applications

Nanomaterials: Introduction, preparation by sol-gel & chemical vapour deposition methods. Applications of nanomaterials.

Suggested Reading:

1. Engineering Chemistry, Jain & Jain
2. Engineering Chemistry, Shashi Chawla
3. Chemistry for Engineers, B. K. Ambasta
4. Engineering Chemistry, H. C. Srivastava

PROGRAMMING FOR PROBLEM SOLVING

Internals: 40 Marks

L - T - P - C

Externals: 60 Marks

3 - 0 - 0 - 3

Objectives: The course will enable the students

- To formulate simple algorithms for arithmetic and logical problems
- To translate the algorithms to programs (in C language)
- To test and execute the programs and correct syntax and logical errors
- To implement conditional branching, iteration and recursion
- To decompose a problem into functions and synthesize a complete program using divide and conquer approach

UNIT-I: Introduction to Programming & Arithmetic expressions and precedence, Introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.). **Idea of Algorithm:** steps to solve logical and numerical problems. **Representation of Algorithm:** Flowchart/Pseudocode with examples. From algorithms to programs; source code, variables (with data types) variables and memory locations, Syntax and Logical Errors in compilation, object and executable code. Arithmetic expressions and precedence.

UNIT-II: Conditional Branching, Loops & Arrays, Writing and evaluation of conditionals and consequent branching, Iteration and loops, Arrays (1-D, 2-D), Character arrays and Strings.

UNIT-III: Function & Basic Algorithms, Functions (including using built in libraries), Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference, Searching, Basic Sorting Algorithms (Bubble, Insertion and Selection), Finding roots of equations, notion of order of complexity through example programs (no formal definition required)

UNIT-IV: Recursion & Structure, Recursion, as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc. Quick sort or Merge sort, Structures, Defining structures and Array of Structures

UNIT-V: Pointers & File handling, Idea of pointers, Defining pointers, Use of Pointers in self-referential structures, notion of linked list (no implementation) File handling (only if time is available, otherwise should be done as part of the lab)

Outcomes:

Upon completion of this course, the students should be able to

- To use arrays, pointers and structures to formulate algorithms and programs
- To apply programming to solve matrix addition and multiplication problems and searching and sorting problems
- To apply programming to solve simple numerical method problems, namely root finding of function, differentiation of function and simple integration

Suggested Text Books

(i) Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill

(ii) E. Balaguruswamy, Programming in ANSI C, Tata McGraw-Hill

Suggested Reference Books

- (i) Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India

Category: **Engineering Science Course**

Subject Code: **ME1703/ME1802**

Engineering Workshop

Internals: 40 Marks

L - T - P - C

Externals: 60 Marks

0 - 1 - 2 - 2

Course Outcomes: Upon completion of this laboratory course

- Students will be able to fabricate components with their own hands.

List of Experiments:

1. **Fitting** – To produce a Step Fit on the given work piece.
 - To produce a V Fit on the given work piece.
2. **Carpentry** – To produce a Half lap joint on the given wooden work part.
 - To produce a Dove tail joint on the given wooden work part.
3. **House Wiring** – To perform and understand the Series and Parallel wiring connections.

- To perform and understand Staircase and Go down wiring connections.

4. **Tin Smithy** – To produce a Tray from the given sheet metal.

- To produce a Cylinder from the given sheet metal.

5. **Welding** – To practice formation of a Bead on the given work piece.

- To perform a Butt and a Lap joint on the given work piece.

6. **Foundry** – To prepare a Mold cavity using a Single piece pattern.

- To prepare a Mold cavity using a Split piece pattern.

7. **Machining** – To perform a Plain turning operation, Facing operation on the given workpiece.

- To perform a Step and a Taper turning operation on the given workpiece.

8. **Plastic molding** – Demonstration

9. **WIRE EDM, CNC, 3D Printer** – Demonstration

Category: **Engineering Science Course**

Subject Code: **19CS1801**

PROGRAMMING FOR PROBLEM SOLVING LAB

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-4-2

List of experiments:

1. Problem solving using computers (Familiarization with programming environment)
2. Variable types and type conversions (Simple computational problems using arithmetic expressions)
3. Branching and logical expressions (Problems involving if-then-else structures)
4. Loops, while and for loops (Iterative problems e.g., sum of series)
5. 1D Arrays: searching, sorting (1D Array manipulation)
6. 2D arrays and Strings, memory structure (Matrix problems, String operations)
7. Functions, call by value (Simple functions)
8. Recursion, structure of recursive calls (Recursive functions)
9. Pointers, structures and dynamic memory allocation (Pointers and structures)
10. File handling (File operations)

Outcomes:

Upon completion of this lab, the students should be able

- To formulate the algorithms for simple problems
- To translate given algorithms to a working and correct program
- To be able to correct syntax errors as reported by the compilers
- To be able to identify and correct logical errors encountered at run time
- To be able to write iterative as well as recursive programs
- To be able to represent data in arrays, strings and structures and manipulate them through a program
- To be able to declare pointers of different types and use them in defining self referential structures.

Textbooks:

1. Byron Gottfried, Schaum's Outline of Programming with C, McGraw-Hill
2. E. Balaguruswamy, Programming in ANSI C, Tata McGraw-Hill

Reference Books:

1. Brian W. Kernighan and Dennis M. Ritchie, the C Programming Language, Prentice Hall of India

Category: **Basic Science Course**

Subject Code: **19CY1802**

ENGINEERING CHEMISTRY LAB

Internals: 40 Marks

Externals: 60 Marks

L - T - P - C

0 - 0 - 3 – 1.5

List of Experiments:

- Thin layer chromatography
- Ion exchange column for removal of hardness of water
- Determination of chloride content of water
- Preparation of Thiokol rubber
- pH Metry
- Synthesis of a drug-Aspirin
- Saponification/acid value of an oil
- Estimation of hardness of given water sample
- Estimation of alkalinity of given sample

Outcomes:

The chemistry laboratory course will consist of experiments illustrating the principles of chemistry relevant to the study of science and engineering. The students will learn to

- Estimate pH, chloride content and alkalinity of given water samples.
- TLC and Ion exchange columns for removal of hardness of water
- Synthesize a small drug molecule, soap and polymer

Category: **Professional Core Course**

Subject Code: **19CE2102**

BUILDING MATERIALS AND CONCRETE TECHNOLOGY

Externals: 60Marks

Internals: 40Marks

L-T-P-C

3-0-0-3

Objectives:

- To study about the basic building materials, properties and their applications
- To know the smart building materials, external paints and their uses.
- To understand different types of masonries and their applications.
- To study about standard dimensions of doors, windows, ventilators and other components of buildings.
- To understand the basic concept of planning of buildings for different storey height.

UNIT – I (8hours)

Stones: Uses of stones as building materials, classification, characteristics, dressing and polishing of stones, methods of quarrying and construction.

Bricks: Methods of manufacturing bricks. Classification and methods of construction.

Timber: Timber as a building material and its uses. Methods of seasoning and preservation laminates and their uses. Scaffolding and form work. Defects in Timber.

Masonry: Brick Masonry: Definition, bonds in brick masonry, principles in brick masonry construction. Stone masonry and basic comparison with brick masonry.

Plastering and pointing: Different types of plasters and plastering process, defects in plastering.

Floors: Common types of floors, methods of construction.

UNIT – II (10hours)

Paints, Varnish and Distemper: Constituents, characteristics of good paints bases, vehicles, thinners and coloring pigments. Painting of different types of surfaces, varnish and its types, application. Distemper, dry and oil bound and application of distemper.

Smart building Materials: Energy conservation in buildings -use of recycled materials, regional materials and industrial waste products as means of sustainable development. Green Building Materials.

Cement: Introduction to cement, different grades, Types of cements and their composition – manufacture of Portland cement – hydration of cement and hydration product, Structure of hydrated cement – heat of hydration, Gel theories and tests on properties of cements.

Mortar and Sand: Characteristics of good mortar making sand, availability of sand and its classification, bulking of sand, manufacturing methods of mortar. Different types of mortars preparation, setting and curing.

UNIT – III (10hours)

Coarse and fine Aggregate: Characteristics of good coarse and fine aggregates for manufacturing of concrete. Classification of aggregates, particle shape and texture, bond strength of aggregates and its influence on strength of concrete, porosity, absorption and moisture content and their influence, soundness of aggregate, alkali aggregate reaction, sieve analysis and grading of aggregate, tests on properties of aggregates.

Concrete: Introduction to concrete, nominal mix and design mix of concrete.

Fresh Concrete: Mixing and batching, workability, factors effecting workability, various test producers, segregation and bleeding, vibration of concrete, types of vibrators and their influence on composition, analysis of fresh concrete.

UNIT – IV (10hours)

Hardened Concrete: strength of concrete, water cement ratio, Gel space ratio, effective water in the mix, short term and long term properties of concrete, test and procedure, influence of various parameters on strength of concrete, relationship between various mechanical strengths of concrete, curing of concrete, maturity concept, influence of temperature on strength of concrete, stress-strain curves for concrete, durability of concrete.

Strength: - Shrinkage and temperature effects - creep of concrete – permeability of concrete – durability of concrete – corrosion – causes and effects – remedial measures – thermal properties of concrete – micro cracking of concrete.

UNIT – V (8hours)

Admixtures: Classification of admixtures, Chemical and mineral admixtures. Influence of various admixtures on properties of concrete. Applications, concept of ready mix concrete- self compacting concrete, fly ash concrete- properties and proportions of fly ash, application, silica fume, rice husk ash concrete

Mix Design: Basic consideration, process of mix design, factors influencing mix proportions-mix design by ACI method and IS code method, design of high strength concrete, quality control, various methods of mix design, IS code method, British and ACI methods.

Outcomes:

Upon completion of this course, the students should be able to

- Categorize and identify different building materials, properties and their uses.
- Recognize the effects of the rheology and early age properties of concrete on its long-term behavior.
- Apply the use of various admixtures to design cement based materials with tailor- made properties.
- Use advanced laboratory techniques to characterize cement-based materials.
- Understand different types of concretes, their engineering properties and usefulness in day to day life.
- Perform the mix design work for the In-situ working conditions

Text/Reference Books:

1. V.N Vazirani, and S.P.Chandola, Engineering Materials, Khanna Publishers 1993
2. Sushil Kumar, Building Construction, Standard Publishers 1992
3. S.P Arora and S.P Bindra, Text book on Building Construction, Dhanapath Raj Publications,1999

4. Dr. N.KumaraSwamy, A.KameswaraRao, Building Planning and Drawing, Charotar Publishers.

Category: **Professional Core Course**

Subject Code: **19CE2103**

MECHANICS OF MATERIALS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-1-0-4

Objectives:

- To understand the basic concept of the stress and strain for different materials
- To know the mechanism of the development of shear force and bending moments in beams
- To analyze and understand bending stress and shear stress
- To study compound stresses, direct and bending stresses in different members
- To know the concept of unsymmetrical bending and shear centre for different members

UNIT-I:(15 hours)

Simple Stresses and Strains: Definition of stresses and strains-Hooke's Law – Modulus of Elasticity -Stress – Strain curve for ductile materials -working stress and factor of safety - Deformation of bars under axial loads -uniform sections and abruptly varying sections - deformation due to self-weight-Impact load, Gradually applied load-Strain energy- Bars of uniform strength -Poisson's ratio-volumetric strain and restricted strains- relationships between elastic constants. Compound bars and temperature stresses: Statically intermediate problems in tensions and compression.

Analysis of stresses: Stress at a point, Components of stress in Rectangular coordinates, Stress tensor, Plane stress, Principal stresses, Mohr's circle for plane stress

UNIT-II: (10 hours)

Thin Cylinders: Thin cylinders subjected to internal fluid pressure, volumetric change.

Thick Cylinders: Lamé's equations, stresses under internal and external fluid pressures-Compound cylinders- Shrink fit pressure.

Torsion: Geometry of deformation of a twisted circular shaft, Stress and deformation in twisted circular solid and hollow shafts, Strain energy due to torsion, Power transmitted by circular shafts.

Springs: Types of springs. Close coiled and open coiled springs.

UNIT-III: (15 hours)

Column stability: Euler's theory for long columns- different end conditions- equivalent length- Rankine's theory. Eccentrically loaded columns- Secant and Perry's formulae.

Shear Force and Bending Moment diagrams: Different types of beams and loads-shear force and bending moment diagrams for cantilever, and simply supported beams with and without overhangs subjected to different kinds of loads viz., point loads, uniformly distributed loads, uniformly varying loads and couples -Relation between loading, shear force and bending moments.

UNIT-IV:(10 hours)

Bending Stresses in Beams: Assumptions in theory of simple bending- Derivation of bending equation, Moment of resistance-calculation of stresses in statically determinate beams for different loads and different types of structural sections.

Shear Stress in Beams: Equation of shear stresses, distribution across rectangular, circular, triangular, L, T sections.

Combined Stresses: Beam subjected to bending and shear, shaft subjected to bending and torsion

UNIT-V:(10 hours)

Deflection in beams: Slope and deflection by double integration method for cantilever, simply supported beams and overhanging beams carrying one, two point loads, uniformly distributed load and uniformly varying load over entire span. Moment area and conjugate beam method.

Course Outcomes:

At the end of the course, the students will have

- Thorough understanding of the fundamental concepts of stress and strain in mechanics of solids and structures
- The ability to analyse determinate beams and trusses to determine shear forces, bending moments and axial forces.
- A sufficient knowledge in designing shaft to transmit required power and also springs for its maximum energy storage capacities
- Understanding of column buckling and ability to calculate critical load and stresses.

Suggested References:

1. Beer, F.P., and Johnston, JR, E.R., "Mechanics of Materials", (2nd Edition), McGraw Hill, 1992.
2. Boresi, A.P., Schmidt, R.J. and Sidebottom, O.M., Advanced Mechanics of Materials (4th Edition), John Wiley and Sons, 1993.
3. Popov, E.P., Mechanics of Materials, Prentice Hall of India Private Limited, 1976.
4. Pytel, A. and Singer, F.L., Strength of Materials (4th Edition), Harper Collins Publishers, Singapore, 1987.
5. Srinath, I.S., Desayi, P.Murthy, N.S. and AnanthaRamu, S., Strength of Materials McMillan India Ltd., 1997.
6. Timoshenko, S. and Young, D.M., Element of Strength of Materials (5th Edition),

FLUID MECHANICS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-1-0-4

Objectives:

- To know various fluid properties, concept and method of fluid pressure measurement.
- To understand the basic concepts of fluid motion.
- To study different equations of fluid motion and fluid dynamics.
- To analyse different flow characteristics of laminar and turbulent flows.
- To study the motion of gases for different conditions of expansion.

Unit - I

Basic concepts and Definitions: Distinction between a fluid and a solid; Density, Specific weight, specific gravity, kinematic and dynamic viscosity; Newton law of viscosity; vapour pressure, boiling point, cavitation; surface tension, capillarity, Bulk modulus elasticity, compressibility.

Fluid Statics- Fluid Pressure: pressure at a point, Pascal's law, pressure variation in a fluid at rest, Absolute, Gauge, Atmospheric and Vacuum pressures, measurement of pressure, simple manometers and Differential manometers. Hydrostatic pressure and force: Horizontal, vertical, inclined surfaces and curved surfaces. Buoyancy and stability of floating bodies.

Unit - II

Fluid Kinematics: Stream lines, path lines, streak lines, stream tubes, classification of fluid flow: steady and unsteady flows, rotational flows, laminar and turbulent flows, uniform and non-uniform flows, rotational and irrotational flows-one, two, three dimensional flows, stream function, and velocity potential function. Convective and local acceleration, concept of continuity, three dimensional continuity equation in Cartesian coordinates.

Unit - III

Fluid Dynamics: Body forces and surface forces; Equation of motion - Euler's equation; Bernoulli's equation – derivation; Energy Principle; Practical applications of Bernoulli's equation: Venturimeter, orifice meter and pitot tube; Momentum principle; Forces exerted by fluid flow on pipe bend; Vortex Flow – free and forced. measurement of flow in open channels, weirs and notches.

Unit – IV

Laminar and Turbulent flow through Pipes: laminar flow through circular pipes, laminar flow between parallel plates, Hazen-Poiseuille equation, Reynolds experiment, Transition from laminar to turbulent flow, head loss in flow through pipes, Darcy Weisbach equation, losses in pipe transitions, Turbulence, Prandtl's mixing length theory, velocity distribution in turbulent flow, resistance to flow of fluid in smooth and rough pipes, Moody's diagram. Total energy equation, hydraulic gradient line, pipes in series, equivalent pipes, pipes in parallel pipes, Analysis of pipe networks: Hardy Cross method, water hammer in pipe and control measures, branching of pipes, three reservoir problem.

Unit - V

Compressible flow: Compressibility of liquid and gases, different form of continuity equation; Bernoulli's energy equation for Isothermal and adiabatic conditions, velocity of pressure wave, wave velocity for Adiabatic and Isothermal conditions, Mach number and Mach cone, Stagnation pressure and temperature.

Introduction to boundary layer theory, necessity, various thicknesses.

Course Outcomes (COs): Upon completion of this course, students should be able to:

- Understand the broad principles of fluid statics, kinematics and dynamics
- Understand definitions of the basic terms used in fluid mechanics and characteristics of fluids and its flow
- Understand classifications of fluid flow
- Be able to apply the continuity, momentum and energy principles

Text Books

1. Fluid Mechanics by Modi and Seth, Standard Book House.
2. Fluid Mechanics and Hydraulic machines by Manish Kumar Goyal, PHI learning Private Limited, 2015.

References

1. Fluid Mechanics by R.C. Hibbeler, Pearson India Education Services Pvt. Ltd. Reference Books
2. Theory and Application of Fluid Mechanics, K. Subramanya, Tata McGraw Hill
3. Introduction to Fluid Mechanics and Fluid Machines by SK Som, Gautam Biswas, Suman Chakraborty, Mc Graw Hill Education (India) Private Limited
4. Fluid Mechanics and Machinery, C.S.P. Ojha, R. Berndtsson and P.N. Chadramouli, Oxford University Press, 2010.
5. Fluid Mechanics and Hydraulic Machines, R.K. Bansal, Laxmi Publication Pvt Ltd.
6. Fluid mechanics & Hydraulic Machines, Domkundwar & Domkundwar Dhanpat Rai & Co

PROBABILITY AND STATISTICS

Internals: 40 Marks

Externals: 60 Marks

L - T - P - C

3- 1 - 0 - 4

Objective:

To make the student to understand the statistics and probability theories such as random variables, sampling distribution, tests of significance etc. so that he can apply them to engineering problems.

UNIT-I:

Probability: Sample space and events – Probability – The axioms of probability – Some Elementary theorems – Conditional probability – Bayes theorem, Random variables – Discrete and continuous.

UNIT-II:

Single Random variables and probability distributions: Random variables- Discrete and continuous. Probability distributions, mass function/ density function of a probability distribution. Mathematical Expectation, Moment about origin, Central moments Moment generating function of probability distribution.

Binomial, Poisson & normal distributions and their properties Moment generating functions of the above three distributions and hence finding the mean and variance.

UNIT III:

Multiple Random variables, Correlation&Regression : Joint probability distributions- Joint probability mass / density function, Marginal probability mass / density functions, Covariance of two random variables, Correlation -Coefficient of correlation, The rank correlation.

Regression- Regression Coefficient, The lines of regression and multiple correlation & regression.

UNIT-IV:

Sampling Distributions and Testing of Hypothesis

Sampling:Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance.

Parameter estimations – likelihood estimate, interval estimations.

Testing of hypothesis: Null hypothesis, Alternate hypothesis, type I, & type II errors – critical region, confidence interval, Level of significance. One sided test, Two sided test,

Large sample tests:

- i) Test of Equality of means of two samples equality of sample mean and population mean (cases of known variance& unknown variance, equal and unequal variances)
- ii) Tests of significance of difference between sample S.D and population S.D.
- iii)Tests of significance difference between sample proportion and population proportion&difference between two sample proportions.

Small sample tests: Student t-distribution, its properties; Test of significance difference between sample mean and population mean; difference between means of two small samples
Snedecor's F- distribution and its properties. Test of equality of two population variances
Chi-square distribution, its properties, Chi-square test of goodness of fit

UNIT- V:

Queuing Theory & Stochastic Processes: Arrival Theorem – Pure Birth process and Death Process M/M/1 Model. Introduction to Stochastic Processes – Markov process classification of states – Examples of Markov Chains, Stochastic Matrix, limiting probabilities.

Outcome: Students will be able to perform probability theories & statistics on engineering problems

Text Books:

- 1) Fundamentals of Mathematical Statistics by S C Gupta and V.K.Kapoor
 - 2) Probability and Statistics for Engineering and The Sciences by Jay L.Devore.
 - 3) Mathematics for Engineers Series –Probability Statistics and Stochastic Process by K.BDatta and M.A S.Srinivas, Cengage Publications
 - 4) Probability, Statistics and Stochastic Process by Prof.A R K Prasad., Wiley India
- A Text Book of Probability and Statistics, Shahnaz Bathul, Cengage Learning

Subject Code: 19CE2205

CIVIL ENGINEERING SOCIETIAL AND GLOBAL IMPACT

Externals: 60Marks

Internals: 40Marks

L-T-P-C

2-0-0-2

Course Objective: The course is designed to provide a better understanding of the impact which Civil Engineering has on the Society at large and on the global arena. Civil Engineering projects have an impact on the Infrastructure, Energy consumption and generation, Sustainability of the Environment, Aesthetics of the environment, Employment creation, Contribution to the GDP, and on a more perceptible level, the Quality of Life. It is important for the civil engineers to realise the impact which this field has and take appropriate precautions to ensure that the impact is not adverse but beneficial.

Course Contents:

Module 1:

Introduction to Course and Overview: (4 Hours)

Understanding the past to look into the future: Pre-industrial revolution days, Agricultural revolution, first and second industrial revolutions, IT revolution; Recent major Civil Engineering breakthroughs and innovations; Present day world and future projections, Evaluating future requirements for various resources; GIS and applications for monitoring systems; Human Development Index and Ecological Footprint of India Vs other countries and analysis;

Module 2:

Understanding the importance of Civil Engineering in shaping & impacting the world: (4 Hours)

The ancient and modern Marvels and Wonders in the field of Civil Engineering; Future Vision for Civil Engineering

Module 3:

Infrastructure - Habitats, Megacities, Smart Cities, futuristic visions: (3 Hours)

Transportation (Roads, Railways & Metros, Airports, Seaports, River ways, Sea canals, Tunnels (below ground, under water); Futuristic systems (ex, Hyper Loop)); Energy generation (Hydro, Solar (Photovoltaic, Solar Chimney), Wind, Wave, Tidal, Geothermal, Thermal energy); Water provisioning; Telecommunication needs (towers, above-ground and underground cabling).

Module 4:

Environment: (3 Hours)

Traditional & futuristic methods; Solid waste management, Water purification, Wastewater treatment & Recycling, Hazardous waste treatment; Flood control (Dams, Canals, River interlinking), Multi-purpose water projects, Atmospheric pollution;

Module 5:

Built environment: (3 Hours)

Recycling, Temperature/ Sound control in built environment, Conservation, Repairs & Rehabilitation of Structures & Heritage structures; Innovations and methodologies for ensuring Sustainability Module 6: Civil Engineering Projects: (3 Hours) Environmental Impact Analysis procedures; Waste (materials, manpower, equipment) avoidance/ Efficiency increase; Advanced construction techniques for better sustainability; Techniques for reduction of Green House Gas emissions in various aspects of Civil Engineering Projects; contribution of Civil Engineering to GDP. Course Outcomes: • The impact which Civil Engineering projects have on the Society at large and on the global arena and using resources efficiently and effectively. • The extent of Infrastructure, its requirements for energy and how they are met: past, present and future • The Sustainability of the Environment, including its Aesthetics, • The potentials of Civil Engineering for Employment creation and its Contribution to the GDP • The Built Environment and factors impacting the Quality of Life

Text or reference book:

- Ashley R M., Nowell R., Gersonius B., Walker L. (2011). Surface Water Management and Urban Green Infrastructure. Review of Current Knowledge. Foundation for Water Research FR/R0014.
- Barry M. (2003) Corporate social responsibility – unworkable paradox or sustainable paradigm? Proc ICE Engineering Sustainability 156. Sept Issue ES3 paper 13550. p 129-130
- Blackmore J M., Plant R A J. (2008). Risk and resilience to enhance sustainability with application to urban water systems. J. Water Resources Planning and Management. ASCE. Vol. 134, No. 3, May.
- Bogle D. (2010) UK's engineering Council guidance on sustainability. Proc ICE Engineering Sustainability 163. June Issue ES2 p61-63.
- Brown R R., Ashley R M., Farrelly M. (2011). Political and Professional Agency Entrapment: An Agenda for Urban Water Research. Water Resources Management. Vol. 23, No.4. European Water Resources Association (EWRA) ISSN 0920-4741.
- Brugnach M., Dewulf A., Pahl-Wostl C., Taillieu T. (2008) Toward a relational concept of uncertainty: about knowing too little, knowing too differently and accepting not to know. Ecology and Society 13 (2): 30
- Butler D., Davies J. (2011). Urban Drainage. Spon. 3rd Ed.
- Cavill S., Sohail M. (2003) Accountability in the provision of urban services. Proc. ICE. Municipal Engineer 156. Issue ME4 paper 13445, p235-244.

Category: **Mandatory Course**

Subject Code: **19BS2102**

ENVIRONMENTAL SCIENCE

Internals: 40 Marks
Externals: 60 Marks

L - T - P - C
2 - 0 - 0 - 2

Vision:

The importance of environmental science and environmental studies cannot be disputed. The need for sustainable development is a key to the future of mankind. Continuing problems of pollution, loss of forest, solid waste disposal, degradation of environment, issues like economic productivity and national security, Global warming, the depletion of ozone layer and loss of biodiversity have made everyone aware of environmental issues. The United Nations Conference on Environment and development held in Rio de Janeiro in 1992 and world Summit on Sustainable Development at Johannesburg in 2002 have drawn the attention of people around the globe to the deteriorating condition of our environment. It is clear that no citizen of the earth can afford to be ignorant of environment issues. Environmental management has captured the attention of health care managers. Managing environmental hazards has become very important.

Human beings have been interested in ecology since the beginning of civilization. Even our ancient scriptures have emphasized about practices and values of environmental conservation. It is now even more critical than ever before for mankind as a whole to have a clear understanding of environmental concerns and to follow sustainable development practices. India is rich in biodiversity, which provides various resources for people. It is also basis for biotechnology. Only about 1.7 million living organisms have been described and named globally. Still many more remain to be identified and described. Attempts are made to conserve them in ex-situ and in-situ situations. Intellectual property rights (IPRs) have become important in a biodiversity-rich country like India to protect microbes, plants and animals that have useful genetic properties. Destruction of habitats, over-use of energy resource and environmental pollution has been found to be responsible for the loss of a large number of life-forms. It is feared that a large proportion of life on earth may get wiped out in the near future.

In spite of the deteriorating status of the environment, study of environment has so far not received adequate attention in our academic programmes. Recognizing this, the Hon'ble Supreme Court directed the UGC to introduce a basic course on environment at every level in college education. Accordingly, the matter was considered by UGC and it was decided that a six months compulsory core module course in environmental studies may be prepared and compulsorily implemented in all the University/Colleges of India. The expert committee appointed by the UGC has looked into all the pertinent questions, issues and other relevant matters. This was followed by framing of the core module syllabus for environmental studies for undergraduate courses of all branches of Higher Education. We are deeply conscious that there are bound to be gaps between the ideal and real. Genuine endeavour is required to minimize the gaps by intellectual and material inputs. The success of this course will depend on the initiative and drive of the teachers and the receptive students.

SYLLABUS:

UNIT 1: Multidisciplinary Nature of Environmental Studies

Definition, scope and importance, need for public awareness.

UNIT 2: Natural Resources

Renewable and non-renewable resources: Natural resources and associated problems.

a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forest and tribal people.

b) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies.

c) Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources.

- Role of an individual in conservation of natural resources.
- Equitable use of resources for sustainable lifestyles.

UNIT 3: Ecosystems & Biodiversity

Concept of an ecosystem. Structure and function of an ecosystem. Producers, consumers and decomposers. Energy flow in the ecosystem. Ecological succession. Food chains, food webs and ecological pyramids.

Introduction, types, characteristic features, structure and function of the following ecosystems: -

- Forest ecosystem, b. Grassland ecosystem, c. Desert ecosystem, d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries).
- Biodiversity- Definition: genetic, species and ecosystem diversity. Biogeographical classification of India Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values.
- Biodiversity at global, National and local levels. India as a mega-diversity nation Hot-spots of biodiversity.
- Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT 4: Environmental Pollution

Definition, Cause, effects and control measures of: - Air pollution, GPM model, Noise pollution, Thermal pollution, and nuclear hazards

- Role of an individual in prevention of pollution
- Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies.
- Environment Protection Act., Air (Prevention and Control of Pollution) Act. Water Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act.

UNIT 5: Social Issues & The Environment

Human Rights. Value Education. HIV/AIDS. Women and Child Welfare. Role of Information Technology in Environment and human health.

Field work: Visit to a local area to document t environmental assets river/forest/grassland/hill/mountain Visit to a local polluted site Urban/Rural/Industrial/Agricultural. Study of common plants, insects, birds. Study of simple ecosystems-pond, river, hill slopes, etc.

REFERENCES:

- a). Agarwal, K.C. 2001 Environmental Biology, Nidi Publ. Ltd. Bikaner.
- b). Bharucha Erach, The Biodiversity of India, Mapin Publishing Pvt. Ltd., Ahmedabad- 380 013, India, Email:mapin@icenet.net (R)
- c). Brunner R.C., 1989, Hazardous Waste Incineration, McGraw Hill Inc. 480p
- d) Clark R.S., Marine Pollution, Clanderson Press Oxford (TB)
- e). Cunningham, W.P. Cooper, T.H. Gorhani, E & Hepworth, M.T. 2001, Environmental Encyclopedia, Jaico Publ. House, Mumabai, 1196p
- f). De A.K., Environmental Chemistry, Wiley Eastern Ltd.
- g). Down to Earth, Centre for Science and Environment (R)
- h). Gleick, H.P. 1993. Water in Crisis, Pacific Institute for Studies in Dev., Environment & Security. Stockholm Env. Institute Oxford Univ. Press. 473p
- i). Hawkins R.E., Encyclopedia of Indian Natural History, Bombay Natural History Society, Bombay (R) j) Heywood, V.H & Waston, R.T. 1995. Global Biodiversity Assessment. Cambridge Univ. Press 1140p.
- k). Jadhav, H & Bhosale, V.M. 1995. Environmental Protection and Laws. Himalaya Pub. House, Delhi 284 p.
- l). Mckinney, M.L. & School, R.M. 1996. Environmental Science Systems & Solutions, Web enhanced edition. 639p.

- m). Mhaskar A.K., Matter Hazardous, Techno-Science Publication (TB)
- n). Miller T.G. Jr. Environmental Science, Wadsworth Publishing Co. (TB)
- o). Odum, E.P. 1971. Fundamentals of Ecology. W.B. Saunders Co. USA, 574p
- p). Rao M N. & Datta, A.K. 1987. Waste Water treatment. Oxford & IBH Publ. Co. Pvt. Ltd. 345p.
- q). Sharma B.K., 2001. Environmental Chemistry. Geol Publ. House, Meerut
- r). Survey of the Environment, The Hindu (M)
- s). Townsend C., Harper J, and Michael Begon, Essentials of Ecology, Blackwell Science.
- t). Trivedi R.K., Handbook of Environmental Laws, Rules Guidelines, Compliances and Standards, Vol I and II, Enviro Media (R).
- u). Trivedi R. K. and P.K. Goel, Introduction to air pollution, Techno-Science Publication (TB).
- v). Wanger K.D., 1998 Environmental Management. W.B. Saunders Co. Philadelphia, USA 499p
- (M) Magazine (R) Reference (TB) Textbook

Category: **Professional Core Course**

Subject Code: **19CE2701**

FLUID MECHANICS LAB

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-2-1

Objectives:

- To identify the behavior of analytical models introduced in lecture to the actual behavior of real fluid flows.
- To explain the standard measurement techniques of fluid mechanics and their applications.
- To analyze the laboratory measurements and to document the results in an appropriate format.

List of experiments:

1. Determination of coefficient of discharge using venturimeter (or) orificemeter.
2. Determination of head loss due to friction in pipes.
3. Verification of Bernoulli's equation.
4. Calibration of Rotameter.

5. Measurement of point velocity using Pitot tube.
6. Determination of the Coefficient of Velocity from Jet Trajectory of small orifices.
7. Determination of the Coefficient of Discharge under Constant Head & Varying head through orifice.
8. Determination of the surface profile of free and forced vortex and comparison with theoretical values.
9. To demonstrate the phenomenon of pipe surge resulting from a change in velocity of the water flowing through a pipe.
10. Determination of the characteristics of water hammer.

Outcomes:

Students who successfully complete this course will have demonstrated ability to:

- Describe the basic measurement techniques of fluid mechanics and its appropriate application.
- Interpret the results obtained in the laboratory for various experiments.
- Compare the results of analytical models introduced in lecture to the actual behavior of real fluid flows and draw correct and sustainable conclusions.

Category: **Professional Core Course**

Subject Code: **19CE2703**

MATERIALS TESTING LAB

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-2-1

Objectives:

- To supplement the theoretical knowledge gained in subject with practical testing by understanding the behavior of materials under different types of loading for different types of structures.
- To enable the student to have a clear understanding of the design for strength and stiffness

List of Experiments:

1. To study the stress strain characteristics of mild steel by Universal Testing Machine.

2. Determination of ultimate shear strength of mild steel specimen by single shear test.
3. Determination of ultimate shear strength of mild steel specimen by double shear test.
4. To determine the hardness of a given material by Rockwell hardness testing machine
5. To determine the hardness of a given material by Brinell's hardness testing machine
6. Determination of Young's modulus using simply supported beam setup.
7. Determination of the energy absorbed and impact strength of steel using Charpy impact testing machine
8. Determination of the energy absorbed and impact strength of steel using Izod impact testing machine.
9. To find the modulus of rigidity of a given specimen by Torsion testing machine.
10. To determine the Shear Modulus (or) Modulus of Rigidity of the given spring material.

Course Outcomes: At the end of the course, the students will be able to

- Conduct tension test on materials like steel etc.
- Conduct compression test on spring, wood, and concrete
- Conduct flexural and torsion test to determine elastic constants
- Determine hardness of metals

Category: **Professional Core Course**

Subject Code: **19CE2702**

CONCRETE TECHNOLOGY LABORATORY

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-2-1

List of experiments:

A) Tests on cement and mortar

1. Fineness of cement by sieving
2. Fineness of cement by air permeability test
3. Consistency of cement
4. Initial setting time and final setting time

5. Preparation and testing of mortar cubes
6. Soundness of cement by using lechartlier's equipment
7. Specific gravity of cement
8. Flow test on mortar

B) Test on aggregates

9. Fineness Chapters of coarse and fine aggregate
10. Bulking of sand
11. Determination of Impurities in sand

C) Tests on concrete for workability

12. Slump test
13. Compaction factor test
14. Vee- bee consistency test
15. V funnel test for self-compacting cement (demonstration)

D) Tests on concrete (strength)

16. Compressive strength
17. Flexural strength (indirect tensile strength test)
18. Rebound hammer test
19. Ultrasonic Pulse velocity test

Outcomes:

At the end of the laboratory, the student will be able to

- Test the various materials as per IS code specification used in construction.
- Judge the suitability of various materials for construction purposes.

Technical Seminar-I

HYDRAULICS ENGINEERING

Externals: 60Marks

Internals: 40Marks

L-T-P-C

3-1-0-4

Objectives:

- To study the concept of the flow through channels and economical design of channels.
- To know the basic concept of gradually varied flow, hydraulic jump and their applications
- To understand the boundary layer theory, concept of drag, lift of streamlined bodies
- To know different theories on dimensional analysis and model concept
- To understand the basic principles of hydraulic turbines, pumps and their hydraulic design

UNIT –I: (15hours)

Steady uniform flow through open channels: Descriptions and definitions, difference between pipe flow and channel flow, Classification of open channels and flows, velocity and pressure distributions in a channel cross section, energy and momentum correction coefficients, friction to flow in open channels, Uniform flow – Characteristics of uniform flow, Chezy's, Manning's and Bazin formulae for uniform flow – Factors affecting Manning's Roughness Coefficient "n". Most efficient channel section, Computation of Uniform flow, Normal depth. Critical Flow: Specific energy, specific energy curve, critical flow, discharge curve specific force specific depth, and critical depth. Channel transitions.

UNI T-II: (14 hours)

Gradually varied flow: Dynamic equation of gradually varied flow, Classification of channel bottom slopes, Classification of surface profile, Characteristics of surface profile, Computation of flow profile by direct step method.

Hydraulic jump: Theory of hydraulic jump, Elements and characteristics of hydraulic jump in a rectangular channel, length and height of jump, location of jump, types, applications and location of jump. Energy dissipation and other uses, surge as a moving hydraulic jump. Positive and negative surges.

UNI T-III: (15 hours)

Dimensional analysis and model studies: Dimensional analysis as a tool in experimental hydraulics, Buckingham's theorem, applications, geometric, kinematic and dynamic Similarity laws, significance of Reynolds, Froude and Mach similarity laws, different types of models and their scale ratios.

Basics of Turbo Machinery: Hydrodynamic force of jets on stationary and moving flat, inclined and curved vanes, Jet striking centrally and at tip, Velocity triangles at inlet and outlet, expressions for work done and efficiency – Angular.

UNI T-IV: (12 hours)

Hydraulic turbines: Elements of a typical Hydropower installation, Heads and efficiencies, Classification of turbines, Pelton wheel, Francis turbine, Kaplan turbine, working, working proportions, velocity diagram, work done and efficiency, hydraulic design. Draft tube Classification, functions and efficiency, characteristic curves.

UNIT- V: (10 hours)

Centrifugal pumps and Reciprocating pumps: Pump installation details, classification, work done, Manometric head, minimum starting speed, losses and efficiencies, specific speed. Multistage pumps, pumps in parallel, performance of pumps, characteristic curves, NPSH, Cavitation.

Outcomes:

At the end of the course the student will able to

- Apply their knowledge of fluid mechanics in addressing problems in open channels and hydraulic machinery.
- Understand and solve problems in uniform, gradually and rapidly varied flows in open channel in steady state conditions.
- Apply dimensional analysis and to differentiate the model, prototype and similitude conditions for practical problems.
- Get the knowledge on different hydraulic machinery devices and its principles that will be utilized in hydropower development and for other practical usages

Text Books

1. Fluid Mechanics by Modi and Seth Standard Book House.
2. Fluid Mechanics and Hydraulic machines by Manish Kumar Goyal, PHI learning Private Limited, 2015
3. Fluid mechanics & Hydraulic Machines, Domkundwar & Domkundwar Dhanpat Rai
4. Fluid Mechanics by R.C. Hibbeler, Pearson India Education Services Pvt. Ltd
5. Fluid Mechanic & Fluid Power Engineering by D.S.Kumar (Kataria & Sons Publications Pvt. Ltd.).
6. Open channel flow by V.T. Chow (McGraw Hill Book Company).
7. Introduction to Fluid Mechanics and Fluid Machines by SK Som, Gautam Biswas, Suman Chakraborty, McGraw Hill Education (India) Private Limited
8. Hydraulic Machines by Banga & Sharma (Khanna Publishers).

Subject Code: **19CE2101**

SURVEYING

Externals: 60Marks

L-T-P-C

Internals: 40Marks

L-T-P-C

3-0-0-3

Objectives:

- ☐ To study the basic concepts and principles of chain survey
- ☐ To know the importance of the compass survey and its practical applications
- ☐ To understand the basic methods and applications of plane table survey
- ☐ To know the field applications and concepts of leveling survey
- ☐ To study the different methods of calculation of area, contouring and measurement of volumes
- ☐ To know the importance of theodolite total station and their practical applications.
- ☐ To study the basic concept of trigonometrical leveling, and field applications.
- ☐ To analyze the horizontal and vertical curves for survey work related to road and railways.
- ☐ To know the principles of aerial photogrammetry and its applications.
- ☐ To study the various applications of GPS, GIS and remote sensing for field work.

UNIT I: (7 hours)

Introduction to Surveying: Principles, classification, Conventional symbols and signals, accessories and phases of surveying.

Linear measurements: Survey stations, Survey lines- ranging, chains and tapes – Errors, corrections,

Compass surveying: Bearing of survey lines, included angles, Dip, Declination & Local Attraction.

UNIT II: (10 hours)

Levelling: Principles of levelling- booking and reducing levels; differential, reciprocal levelling, profile leveling and cross sectioning. Digital and Auto Level, Errors in leveling, Adjustments; Concept of curvature of earth and refraction

Contouring: Characteristics, methods, uses;

Areas: Calculation of areas consisting of Regular & Irregular Boundaries

Volumes: Determination of volume of earth work in cutting and embankments for level sections, Volume of borrow pits, Capacity of Reservoir.

UNIT II: (8 hours)

Triangulation & Trilateration-Theodolite survey: Types of Instruments, Measurement of horizontal and vertical angle; Horizontal and vertical control - methods - triangulation network-Signals. Baseline - choices - instruments and accessories - extension of base lines - corrections - Satellite station - reduction to centre - Intervisibility of height and distances - Trigonometric leveling- Axis single corrections.

UNIT IV: (14 hours)

Curves & Modern Field Survey Systems: Elements of simple and compound curves –Method of setting out–Elements of Reverse curve - Transition curve –Vertical curves Principle of Electronic Distance Measurement, Modulation, Types of EDM instruments, Distomat, Total Station – Parts of a Total Station –Accessories –Advantages and Applications, Field Procedure for total station survey, Errors in Total Station Survey; Global Positioning Systems-Segments, GPS measurements, errors and biases, Surveying with GPS, Co-ordinate transformation, accuracy considerations.

UNIT V: (7 hours)

Photogrammetric survey & Remote Sensing: Introduction to Photogrammetry-Basic Concepts – Aerial photograph – Concept of Scale – Flying Height – Mosaics - mapping; Remote sensing- Definitions – Advantages - Electromagnetic Spectrum, Components of Remote sensing, platforms and sensors; visual image interpretation; Introduction, Basics of GIS.

Outcomes:

Upon completion of this course, the students should be able to

- ☐ Apply the knowledge, techniques, skills, and applicable tools of the

- discipline to engineering and surveying activities
- Translate the knowledge gained for the implementation of Civil infrastructure facilities
- Relate the knowledge on Surveying to the new frontiers of science like Hydrographic Surveying, Electronic Distance Measurement.

Text/Reference Books:

1. Madhu, N, Sathikumar, R and Satheesh Gobi, Advanced Surveying: Total Station, GIS and Remote Sensing, Pearson India, 2006.
2. Manoj, K. Arora and Badjatia, Geomatics Engineering, Nem Chand & Bros, 2011
3. Bhavikatti, S.S., Surveying and Levelling, Vol. I and II, I.K. International, 2010
4. Chandra, A.M., Higher Surveying, Third Edition, New Age International (P) Limited, 2002.
5. Anji Reddy, M., Remote sensing and Geographical information system, B.S. Publications, 2001.
6. Arora, K.R., Surveying, Vol-I, II and III, Standard Book House, 2015.
7. "Surveying (Vol – 1, 2 & 3), by B. C Punmia, Ashok Kumar Jain and Arun Kumar Jain -Laxmi Publications (P) Ltd., New Delhi.
8. Duggal S K, "Surveying (Vol – 1 & 2), Tata McGraw Hill Publishing Co. Ltd. New Delhi, 2004.
9. Surveying and levelling by R. Subramanian, Oxford university press, New Delhi

Category: **Professional Core Course**

Subject Code: **19CE2203**

STRUCTURAL ANALYSIS-I

Externals: 60 Marks

L-T-P-C

Internals: 40 Marks

3-0-0-3

Objectives:

- To solve the statically indeterminate structures by applying the principles of equilibrium and compatibility in deformation pertaining to the structure.
- To analyze the statically determinate arches and find the deflections in beams, frames using energy principles.
- To Draw the Influence line diagrams for reaction, SF and BM
- To evaluate the displacements and redundant forces using energy methods.

UNIT - I: (12 hours)

Slope deflection method: Application of the method to continuous beams with and without sinking of supports, single bay-bay portal frames (Degree of freedom not exceeding three), loading on each span may be point load(s) and /or uniformly distributed load on whole span, shear force and bending moment diagrams.

UNIT - II: (8 hours)

Moment distribution method: Application of the method to continuous beams with and without sinking of supports, single bay-bay portal frames (Degree of freedom not exceeding three), loading on each span may be point load(s) and /or uniformly distributed load on whole span, shear force and bending moment diagrams.

UNIT – III: (8 hours)

Kani's method: Analysis of continuous beams –including settlement of supports and single bay, single storey portal frames with side sway by Kani's method.

Plastic Analysis: Introduction –Idealized stress –Strain diagram –shape factors for various sections –Moment curvature relationship –ultimate moment –Plastic hinge –lower and upper bound theorems –ultimate strength of fixed and continuous beams.

UNIT - IV: (12 hours)

Strain energy method: Deflections of statically determinate trusses and frames using unit load method.

Redundant trusses and frames: Analysis of plane trusses with one degree of redundancy(internal/external) and plane frames with one degree of redundancy, Lack of fit and temperature effect.

UNIT – V: (10 hours)

Three hinged Arches: Elastic theory of arches- Eddy's theorem- Determination of horizontal thrust, bending moment, normal thrust and radial shear for static loading-Effect of temperature

Two hinged Arches: Parabolic and segmental, determination of horizontal thrust, bending moment, normal thrust, and radial shear for static loading-Rib shortening-Effect of temperature.

Outcomes:

Upon completion of this course, the students should be able to

- Understand the advantage of statically indeterminate structure over the statically determinate structure.
- Sketch the SF and BM diagrams for determinate and indeterminate beams.
- Calculate the deflections in beams and pin jointed trusses.
- Evaluate the maximum SF and BM due to various types of moving loads.

Suggested Reading:

1. Structural Analysis by RcHibbeler, 6th Edition, Pearson Publications
2. Basic Structural Analysis - C.S. Reddy, Tata McGraw Hill.
3. Intermediate Structural Analysis - C. K. Wang, McGraw-Hill
4. Theory of Structures - Volumes 1 and 2, S P Gupta and G S Pandit, Tata McGraw Hill.
5. Structural Analysis - L.S.Negi& R.S. Jangid, Tata McGraw Hill.
6. Classical Structural Analysis - Anthony E. Armenikas, McGraw Hill.
7. S.B. Junakar and Shah, Mechanics of structures, Charotar Pub. House, 2001.
8. D.S. PrakashRao, Structural analysis – A Unified Approach, University Press, 1996.
9. B.C. Punmia and A.K. Jain, Theory of structures, Laxmi Publications, New Delhi, 2004.

Category: **Professional Core Course**

Subject Code: **19CE2204**

DESIGN OF CONCRETE STRUCTURES

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To know the IS codal provisions as applicable for the designs.
- To understand the design philosophies and basics of RCC structural designs.
- To understand the design principles in flexure, shear and torsion.
- To learn the design of various components of RCC structures.

UNIT – I: (14hours)

Design Philosophies: Development of design Philosophies-Working stress method, Ultimate load method, and Limit state method – Concepts, Characteristics loads and strengths, Partial safety factors, stress-strain relationship for concrete and steel, stress block parameters.

Working Stress Method: Design of RCC beams – Balanced, under – reinforced and over Reinforced sections – Rectangular, T and L sections, Design of singly and doubly reinforced Rectangular, T and L sections.

UNIT – II: (14hours)

Limit state of collapse in flexure: Assumptions, Design for flexure – Singly and doubly Reinforced rectangular, T and L sections.

Limit state of collapse in shear and torsion: Design for shear and torsion. Limit states of Serviceability: Check for deflection and cracking.

UNIT – III: (14hours)

Design of slabs (Limit state method): Design of one way and two way slabs – Simply supported and continuous slabs subjected to uniformly distributed loads, Detailing of reinforcement, Check for serviceability of slabs.

Design of stair cases (Limit state method): Types of stairs, Effective span, Distribution of Loading on stairs, Design and detailing of dog – legged stair cases.

UNIT – IV: (10hours)

Design of columns (Limit state method): Assumptions, Design of axially loaded circular Square and rectangular columns, Design of columns with Uni–axial and Bi–axial bending Interactions diagrams.

UNIT – V: (8hours)

Design of footings (Limit state method): Design of isolated footings of uniform depth and Sloped footing, Design of square, rectangular and circular footing as per IS code, Design of Combined rectangular slab footing, combined rectangular beam and slab footing for two columns.

Outcomes:

Upon completion of this course, the students should be able to

- Explain the general mechanical behavior of reinforced concrete
- Analyze and design reinforced concrete flexural members
- Design one way, two way slabs and columns
- Design footings and Summarize working stress method

Suggested Reading:

1. IS 456:2000/SP-16 Charts/IS-875:1987 part I-V
2. Punmia B.C., Jain A.K. And Jain A.K., RCC Designs, Laxmi Publications, 2006.
3. Krishna Raju N. And Pranesh R.N., Reinforced Concrete Design, New Age International Pvt, Ltd., 003.
4. Varghese P.C; Limit State Design of Reinforced Concrete, Prentice of India Pvt, Ltd., 2002.
5. Varghese P.C; Design of Reinforced Concrete, Foundations, PHI Learning Pvt. Ltd.,2009.
- 6.D.S. PrakashRao, Design Principles and Detailing of Concrete Structures, Tata McGraw Hill Publishing Co. Ltd., 1995.

CE3101

WATER RESOURCES ENGINEERING - I

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

PreRequisites: Fluid Mechanics & HE

Objectives:

- To introduce the concepts of prediction, preservation, and vision about restoration of dwindling natural resources.
- To impart knowledge about the natural resources, their conservation, and efforts towards their sustainability.
- To understand the utility of Irrigation and crop requirements.

Proposed Syllabus:

Chapter 1: *Introduction (10hours):*

Hydrologic cycle, water-budget equation, history of hydrology, world water balance, applications in engineering, sources of data.

Precipitation:

Forms of precipitation, characteristics of precipitation in India, measurement of precipitation, rain gauge network, mean precipitation over an area, depth-area duration relationships, maximum intensity/depth-duration-frequency relationship, Probable Maximum Precipitation (PMP), rainfall data in India.

Chapter 2: *Abstractions from precipitation (7hours)*

Evaporation process, evaporimeters, analytical methods of evaporation estimation, reservoir evaporation and methods for its reduction,

Evapotranspiration, measurement, equations, potential evapotranspiration, actual evapotranspiration, interception, depression storage,

Infiltration, infiltration capacity, measurement of infiltration, modeling of infiltration capacity, classification of infiltration capacities, infiltration indices.

Chapter 3: *Runoff & Floods (10 hours)*

Runoff volume, SCS-CN method of estimating runoff volume, flow-duration curve, flow-mass curve

Hydrograph, factors affecting runoff hydrograph, components of hydrograph, base flow separation, effective rainfall, Unit hydrograph and its derivation from isolated and complex storms, Concept of S-curve hydrograph, Synthetic Unit Hydrograph, IUH.

Floods:

Flood estimation by different methods, Risk and reliability - flood routing through reservoirs and channels, flood control measures.

Chapter 4: Ground water and well hydrology (6 hours)

Forms of subsurface water, saturated formation, aquifer properties, basic definitions; geologic formations of aquifers, well hydraulics: steady state flow in wells, equilibrium equations for confined and unconfined aquifers, aquifer tests.

Chapter 5: Water withdrawals and uses (Irrigation Engineering) (10 hours)

Irrigation-Necessity, Advantages and disadvantages, Quality of irrigation water; Water requirement of crops-Crops and crop seasons in India, crop period, base period, Basic definitions; cropping pattern, duty and delta; Methods to Improve duty, Soil-water-plant relationships, root zone of soil water, Soil moisture stress, tension, consumptive use, irrigation requirement, frequency of irrigation, Various Irrigation Efficiencies; Methods of applying water to the fields: surface, sub-surface, sprinkler and trickle / drip irrigation.

Text/Reference Books:

1. K Subramanya, Engineering Hydrology, Mc-Graw Hill.
2. K N Muthreja, Applied Hydrology, Tata Mc-Graw Hill.
3. K Subramanya, Water Resources Engineering through Objective Questions, Tata Mc-Graw Hill.
4. G L Asawa, Irrigation Engineering, Wiley Eastern
5. K. N. Duggal, J. P. Soni, Elements of Water Resources Engineering, New Age International Publishers.
6. J D Zimmerman, Irrigation, John Wiley & Sons
7. C S P Ojha, R Berndtsson and P Bhunya, Engineering Hydrology, Oxford.
8. VenteChow, Hand book of applied Hydrology, McGraw-Hill Book Company, New York, 1964
9. H.M Raghunath, Hydrology-Principles, Analysis and Design, New Age International Publishers, 1996.
10. Michael, A.M, Irrigation Theory & Practice, Vikas Publishing House, New Delhi, 1978.
11. Ray K. Linsley, Jr., Max A. Kohler, Joseph L. H. Paulhus, hydrology for Engineers, McGraw-Hill Book Company.

Outcomes:

At the end of the course, students must be in a position to:

- Understand the interaction among various processes in the hydrologic cycle
- Apply the application of fluid mechanics and use of computers in solving a host of problems in hydraulic engineering
- Study types and classes of hydrologic simulation models and design procedures for safe and effective passage of flood flows for design of hydraulic structures
- Understand the basic aquifer parameters and estimate groundwater resources for different hydro-geological boundary conditions.

CONSTITUTION OF INDIA (COI)

Subject Code: **BM1105**

L – T – P –C

Externals: 60 marks

2 – 0 – 0– 0

Internals: 40Marks

Course Out comes: At the end of the course the student will be able to

CO1: Understand the formation and principles of Indian Constitution.

CO2: Understand Fundamental Rights and its implications in life

CO3: Understand Fundamental Duties of Individual toward country and society

CO4: Understand Directive principles to govern the policy formation

CO5: Understand the Way of running the Government and basic Governance

UNIT-I

Introduction to Indian Constitution:

- Meaning of the term Constitution
- Historical background of Indian constitution
- Making of Indian constitution
- Constituent Assembly

• UNIT-II

Features of Indian Constitution

- Preamble of the Constitution , Importance, Scope, Relevance
- The Salient Features of Indian Constitution , Importance, Scope, Relevance

UNIT-III

Fundamental Rights:

- Fundamental Rights
- Importance and scope of fundamental rights
- Categorization of Fundamental Rights

UNIT-IV

Fundamental Duties & The Directive Principles of State Policy:

- Fundamental Duties
- Importance and scope of fundamental Duties
- The Directive Principles of State Policy - Importance, Scope, Relevance

UNIT-V

Union/Central Government: Union Government

- Union Legislature (Parliament)
- Lok Sabha and Rajya Sabha (with Powers and Functions)
- Electronics and Communication Engineering Page 27 Union Executive
- President of India (with Powers and Functions)
- Prime Minister of India (with Powers and Functions)

TextBooks:

1. 'Indian Polity' by Laxmikanth
2. 'Indian Administration' by Subhash Kashyap
3. 'Indian Constitution' by D.D. Basu
4. 'Indian Administration' by Avasti and Avast

Category: **Professional Core Course**

Subject Code: **19CE2801**

HYDRAULICS ENGINEERING LAB

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-2-1

Objectives:

- To illustrate the students with the components and working principles of the Hydraulic machines- different types of Turbines, Pumps, and other miscellaneous hydraulics machines.

- To analyze the laboratory measurements and to document the results in an appropriate format.

List of experiments:

1. Measurement of coefficient of discharge through notches (V notch and Rectangular notch).
2. To investigate the reaction forces produced by the change in momentum of a fluid flow.
3. To demonstrate ground water flow and the resulting hydraulic gradient between two different potentials.
4. To determine the Cone of Depression for a single/double well in an unconfined aquifer.
5. To obtain the characteristic curves for an Impulse turbine operating at a range of fluid flow rates.
6. To obtain the characteristic curves for Reaction turbine operating at a range of fluid flow rates.
7. To obtain the characteristic curves for Francis turbine operating at a range of fluid flow rates.
8. To determine Rainfall-runoff relationships (storm hydrographs), Generation of Overland Flow, sediment yield using Advanced Environmental Hydrology System.
9. Determination of efficiency of multistage Centrifugal pump.
10. Determination of Coefficient of Discharge through Sluice Gate.
11. Determination of Force on a Sluice Gate.
12. Determination of Critical Depth and Derivation of Specific Energy Equation for a Sluice Gate.
13. Determination of Energy loss due to Hydraulic Jump.

Outcomes:

Students who successfully complete this course will have demonstrated ability to:

- Discover the practical working of Hydraulic machines- different types of Turbines, Pumps, and other miscellaneous hydraulics machines.
- Compare the results of analytical models introduced in lecture to the actual behavior of real fluid flows and draw correct and sustainable conclusions.

COMPUTER AIDED DRAFTING OF BUILDINGS LABORATORY

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-3-1.5

Objectives:

- To apply the AutoCAD tools to create building plans, sections and elevations from a given line drawing and specifications.
- To use the Auto cad commands for drawing 2D & 3D building drawings required for different Civil engineering applications.
- To generate the structural drawings of structural elements.
- To plan and draw Civil Engineering Buildings as per aspect and orientation.

List of experiments:

1. Introduction to computer aided drafting and different coordinate system
2. Brick Bonds: English bond & Flemish bond –Odd & even courses for one, one and half, two, two and half brick walls.
3. Exercise on Modify tools & other tools (Layers, dimensions, texting etc.)
4. Drawing of plans of buildings using software
 - a) Single storied buildings
 - b) multi storied buildings
5. Developing sections and elevations for given
 - a) Single storied buildings
 - b) multi storied buildings
6. Detailing of singly reinforced rectangular beam section
7. Detailing of singly reinforced rectangular beam section
8. Detailing of simply supported one way and two way slab
9. Detailing of rectangular column and circular column with spiral stirrups
10. Detailing of isolated square and rectangular footing
11. Detailing of dog legged staircase
12. Detailing of Beam-column junction

Outcomes:

Upon completion of this course, the students should be able to

- Use different AutoCAD Commands to develop Plan, Section and elevation of Buildings.
- Draw and detail different Components of different types of doors and windows.
- Develop Working Drawings of Residential Buildings.
- Prepare structural drawings with details of Beam, Slab and Footing.

TEXT BOOKS

1. Civil Engineering Drawing-I by N. Sreenivasulu, S. Rama Rao –Radiant Publishing house, 2nd edition, 2013.
2. Civil Engineering Drawing II by N. Sreenivasulu – Radiant Publishing house, 2nd edition, 2013.
3. Computer Aided Design Laboratory by M. N. Sesha Praksh & Dr. G. S. Servesh – LaxmiPublications.
4. Engineering Graphics by P. J. Sha – S. Chand & Co.

REFERENCES

1. Engineering Graphics by P. J. Sha- S. Chand & Co, 2nd edition, 2013.
2. Civil Engineering Drawing-I by S. Mahaboob Basha –Falcon Publishers, 2nd edition, 2013.
3. Building drawing by M G Shah -Tata McGraw-Hill Education, 5th edition, 2012
4. Structural Engineering Drawing by S. Mahaboob Basha –Falcon Publishers, 2nd edition, 2013.

Category: **Humanities and Social Sciences including Management Courses**

Subject Code:

19HS2202

EFFECTIVE TECHNICAL COMMUNICATION –I

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-2-1

Module 1: Information Design and Development- Different kinds of technical documents, Information development life cycle, Organization structures, factors affecting information and document design, Strategies for organization, Information design and writing for print and for online media.

Module 2: Technical Writing, Grammar and Editing- Technical writing process, forms of discourse, Writing drafts and revising, Collaborative writing, creating indexes, technical writing style and language. Basics of grammar, study of advanced grammar, editing strategies to achieve appropriate technical style. Introduction to advanced technical communication, Usability, Human factors, Managing technical communication projects, time estimation, Single sourcing, Localization.

Module 3: Self Development and Assessment- Self assessment, Awareness, Perception and Attitudes, Values and belief, Personal goal setting, career planning, Self-esteem.

Outcome:

The student will acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills

Text/Reference Books:

1. David F. Beer and David McMurrey, Guide to writing as an Engineer, John Wiley. New

York, 2004

2. Diane Hacker, Pocket Style Manual, Bedford Publication, New York, 2003. (ISBN 0312406843)

3. Shiv Khera, You Can Win, Macmillan Books, New York, 2003.

AICTE Model Curriculum for Undergraduate degree in Civil Engineering (Engineering & Technology)

4. Raman Sharma, Technical Communications, Oxford Publication, London, 2004.

5. Dale Jungk, Applied Writing for Technicians, McGraw Hill, New York, 2004. (ISBN: 07828357-4)

6. Sharma, R. and Mohan, K. Business Correspondence and Report Writing, TMH New Delhi 2002.

7. Xebec, Presentation Book, TMH New Delhi, 2000. (ISBN 0402213)

Surveying Lab

Technical Seminar-II

Category: **Professional Core Course**

Subject Code: **19CE3101**

ESTIMATION AND COSTING

Externals: 60 Marks

L-T-P-C

Internals: 40 Marks

3-0-0-3

Objectives:

To know the importance of preparing the types of estimates under different conditions

To know about there ate analysis, specifications & bill preparations
To understand the valuation of landand buildings
Tointroducethestudentsindepthknowledgeofprofessionalpracticeaswellthequantityanalysis
ofconstructionworkslike,multi-
storiedstructures,Waterworks&sanitaryworks,Irrigationworks,Roadestimates,culverts,etc.

UNIT-I:(8hours)

GeneralitemsofworkinBuilding-Principlesofworking-
outquantitiesfordetailedandabstractestimates-

ApproximatemethodofEstimationofEarthworkforroadsandcanal

Specifications-GeneralandDetailedspecifications-Typesofestimates-revisedestimates-
Supplementaryandrevisedestimates

UNIT-II:(14hours)

Estimationofbuildings-Detailedestimationofbuildings-TraditionalandAdvancesBuildings-
Longwall&Shortwallmethod-Centerlinemethod-
EstimationofReinforcementquantitiesofbuildingelements-
Beam,Slab,ColumnandColumnwithfooting-Barbendingschedules

UNIT-III:(12 hours)

EstimationofRoadsandcanals-

estimationofroadworksusinglevels(crossectionsandlongitudinalsections)-
irrigationcanalincluingearthwork-estimationofseptictanks-reinforcementquantitiescalculations

UNIT-IV:(8hours)

RateAnalysis-Generalitemsinanalysisofrates-analysisofratesforbuildingandroadworks-
sanitaryandwatersupplyworks

Valuation-**Gross**income,netincome-sinkingfund-salvagevalue-PurposeofValuation-
depreciation-methodsofdepreciation.

UNIT-V:(8hours)

TendersandContracts-typesofcontractsandconditionsofcontracts-contractdocuments-
turnkeycontract-essentialofcontracts-Scheduleofrates-billofquantities-standarddata-
classificationoftenders-basicterminologyintenders-tenderingprocedure-tenderdocumentation

Recordmaintenance: Workabstract,materialatsiteaccount,measurementbook,musterrolehiringand
maintenanceofequipment,recordofbills,vouchersandreceiptbook

Outcomes:

Attheendofthecoursestudentswillbeableto

Understand the procedure to carry out the estimation and steps to prepare detailed reports of construction works.

Prepare quantity estimates for buildings, roads, rails and canal works

Calculate the total quantity of materials required for various civil engineering works as per specifications

Evaluate the procedure of contracts and tenders in construction practice

Suggested Reading:

1. B.N. Dutta, Estimating and Costing in Civil Engineering, Ubs Publishers Distributors Ltd

2. S.C. Rangwala, Estimating and Costing, Charotar Publishing House, Anand

3. G.S. Biridi, Textbook of Estimating & Costing, Dhanapat Rai & Sons, Delhi.

4. Standards schedule of rates and standard data book by public works department

5. I.S.1200 (Parts I to XXV – 1974) Method of measuring building and civil engineering Works-BIS

6. National building code

7. M. Chakroborti, Estimating, Costing, Specification and Valuation, Calcutta.

Category: **Professional Core Course**

Subject Code: **19CE3102**

STRUCTURAL ANALYSIS-II

Externals: 60 Marks

L-T-P-C

Internals: 40 Marks

3-0-0-3

Objectives:

- To understand the analysis of structural elements subjected to moving loads & the analysis of road/railway bridges and gantry girders.
- To understand displacement methods of analysis for statically indeterminate structures.
- To differentiate the various methods of analysis for indeterminate structures.
- To find the degree of static and kinematic indeterminacies of the structures.

UNIT-I: (12 hours)

Moving loads: Influence line for support reaction, bending moment and shear force at any location for simple beams. Determination of maximum bending moment and shear force for moving load system on simply supported girders.

Curves of maximum bending moment and shear force: for simply supported girders traversed by (1) single point load, (2) two point loads, (3) uniformly distributed load longer/shorter than span, enveloping parabola and **EUDLL**.

UNIT-II: (12 hours)

Moving loads on trusses/girders: Influence line for forces in members of statically determinate plane framed structures under moving loads for Warren girder, Pratt truss and Curved flange truss.

Suspension Bridges: Stresses in suspended loaded cables, length of cable, simple suspension bridge with 3-hinged stiffening girders for static load, Influence lines for horizontal and vertical components of tension in the cable, tension in the cable, bending moment and shear force.

UNIT-III: (8 hours)

Flexibility Matrix Method: Determination of static and kinematic indeterminacy – Equilibrium and compatibility conditions-Principles of superposition, Application of Flexibility Matrix Method to continuous beams, plane-trusses, plane-forms and Ortho grid structures (Static indeterminacy not exceeding three)-Effect of temperature, Lack of fit and Pre-stressing forces

UNIT-IV: (8 hours)

Stiffness Matrix Method: Application of Stiffness Matrix Method to continuous beams, plane trusses, plane frames and Ortho grid structures (Degree of freedom not exceeding three) Construction of stiffness Matrix for frames-Direct Method.

UNIT-V: (10 hours)

Direct Element Method: Development of Stiffness matrices for bars, truss and beam elements Application of direct element method to problems of axially loaded bars, continuous beams, plane trusses and plane frames to obtain joint displacements and member end forces. Developing shear force and bending moment diagrams. Instruction to software package **STAAD Pro**

Outcomes:

Upon completion of this course, the students should be able to

- Sketch the shear force and bending moment diagrams for indeterminate structures.
- Formulate the stiffness matrix and analyze the beams by matrix methods.
- Analyze cables and suspension bridges.
- Solve statically indeterminate beams and portal frames using classical method

Suggested Reading:

- 1.S.BJunarkar and Shah, Mechanics of structures, Charotar Pub, House,2001
- 2.D.SPrakashRao, Structural Analysis -a Unified Approach, University Press, 1991
- 3.B.C Punmia and A.K Jain, Theory of structures, Vol.1, Tata McGraw Hill,New Delhi, 1999.

Subject Code: **19CE3103**

HIGHWAY AND TRAFFIC ENGINEERING

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To create overall awareness about highway engineering
- To emphasize the significance of geometric design of highways with specifications and standards
- To create the awareness of different pavement materials and their characteristics
- To impart knowledge on pavement engineering and traffic engineering

UNIT-I (12 hours)

Highway Alignment and Geometric Design: History of highway engineering, factors to be considered for highway alignment, engineering surveys, obligatory points. Geometric design

- Highways classification as per IRC and its standard dimensions, carriageway, shoulders, medians, right of way, footpaths, cycle tracks, service roads, frontage roads, sight distance, stopping sight distance, overtaking sight distance. Camber, horizontal curves, super-elevation, transition curve, extra widening, gradient, grade compensation and design of vertical curves.

UNIT- II (10 hours)

Traffic Engineering: Objectives of traffic studies, traffic characteristics, volume, speed, density, headways and relationship among them. Traffic volume studies, speed and delay studies, intersection delay studies, highway capacity and

level of service concept as per HCM2000, origin and destination studies, concept of signal design, saturation flow, parking and accident studies, highway lighting, problems.

UNIT-III (6 hours)

Intersection Design: Types of Intersections – Conflicts at Intersections- Types of At-Grade Intersections- Channelization: Objectives –Traffic Islands and Design Criteria-Types of Grade Separated Intersections- Rotary Intersection planning – Concept of Rotary and DesignCriteria

UNIT-IV (10 hours)

Highway Materials: Highway materials Characterization-Road aggregate properties- strength, hardness, toughness, abrasion, water absorption, soundness, shape and specific gravity. Bitumen properties - bitumen and tar, grade of bitumen, viscosity grade bitumen, tests on bitumen such as penetration, ductility, viscosity, softening point, flash and fire point. Cutback bitumen. Emulsion, prime coat and tack coat, stripping value of aggregate etc. soil classification, liquid limit and plastic limit, compaction and densities, California bearing ratio value and its significance in pavement design.

UNIT-V (10 hours)

Pavement Design: Pavement types, factors to be considered for pavement design, structural difference between flexible and rigid pavement design. Flexible pavement design - concept of layer theory, design wheel load, ESWL, EALF, vehicle damage factor, design by CBR developed by US corps of engineers, IRC cumulative standard axles method (IRC - 37: 2002). Rigid pavement design - concept, wheel load stresses analysis by Westerguard. Sub-grade, dry lean concrete, radius of relative stiffness, modulus of sub grade reaction and other characteristics of concrete, critical wheel load and temperature stresses. Longitudinal and transverse joints, contraction joints, expansion joints, construction joints, dowel bars and tie bars functions.

Outcomes:

Upon completion of this course, the students should be able to

- Understand the factors influencing road vehicle performance

characteristics and design

- Carry out traffic studies and implement traffic regulations and control measures
- Design basic horizontal alignment and vertical alignment of the highway
- Design flexible and rigid pavements as per IRC

Suggested Reading:

1. Khanna, S. K. and Justo, C. E. G (1994), "Highway Engineering", Nemchand & Bros, New Delhi, India.
2. McShane, W.R., Roess, R.P. and Prassas, E.S., Traffic Engineering. Prentice Hall, Englewood Cliffs, 1997.
3. Highway Capacity Manual, Transportation Research Board, National Research Council. Washington, D.C., 2000.
4. L.R. Kadiyali, *Principles and Practice of Highway Engineering*, Khanna Publications, New Delhi, 2000
5. Haas and Hudson W. R., "Pavement Management Systems" McGraw Hill publications 2004
6. Y.H. Huang, *Pavement Analysis and Design*, Prentice Hall, Englewood Cliffs, New Jersey, 2004

Category: **Professional Core Course**

Subject Code: **19CE3104**

SOIL MECHANICS

Externals: 60 Marks

L-T-P-C

Internals: 40 Marks

3-0-0-3

Objectives:

- To enable characterization of soils based on index and engineering properties.
- To understand soil-water interaction.
- To explain strength and compressibility characteristics of soils.
- To describe the machines involved in earth pressure and slope stability analysis.
- To equip the students for application of principles of soil mechanics in foundation engineering.

UNIT - I

Origin and Characteristics of soil: Introduction, Engineering geology and formation of soil, Weight ratios (Water content, Density, Unit weights, Specific Gravity); Volume ratios (void ratio, porosity, degree of saturation, relative density); Interrelationships, Laboratory tests for

determination of Index properties, Atterberg limits. Classification and Identification of soils for general and engineering purposes as per IS: 1498-1970, Soil structure and Clay minerals.

UNIT - II

Permeability: Bernoulli's equation, Darcy's law for flow through soils, hydraulic conductivity, Factors affecting permeability, Laboratory tests for determination of coefficient of permeability (constant head, variable head permeability tests), Field tests (pumping in and pumping out tests), Equivalent permeability of stratified soils.

Seepage through soils: Seepage, Flow nets, Locating phreatic line in a homogeneous earthen dam, Seepage calculations for geotechnical structures (Masonry dams, concrete dams, earth dams), Uplift pressure.

Stress analysis: In-Situ Stresses, Effective stresses, Pore water pressure, Capillary rise, piping, quicksand condition – critical hydraulic gradient.

UNIT - III

Compaction of soil: Compaction Mechanism; factors affecting compaction. Laboratory determination of compaction characteristics – standard and modified proctor tests – IS Light and Heavy compaction tests; Field surface compaction: compaction equipment, procedure, quality control.

Consolidation of soil: Settlement of Soil, Immediate settlement, Over-consolidation, Pre-consolidation pressure, C_c and C_s , Terzaghi's theory of one dimensional consolidation – assumptions and derivation, Determination of coefficient of consolidation, Calculation of settlements using C_c , m_v and rate of settlement (C_v , T_v , U)

UNIT - IV

Shear Strength of Soils: Stresses on soil elements, Mohr's Circle, Principal Stresses, Pole, Stress Path, Direct shear test, Mohr-Coulomb Failure criterion, Tri-axial tests- drained and undrained tests, Vane shear test, stress-strain relationships and strength of loose and dense sands, NC and OC clays, dilation, pore pressures, Skempton's coefficients, Shear Strength of unsaturated soil, Liquefaction phenomenon.

UNIT - V

Stability of slopes: Definition and classification of slopes, types of slope failure, Factor of safety with respect to cohesion, Analysis of slope stability for finite and infinite slopes - limit equilibrium method, Swedish slip circle method, Taylor's stability number, method of slices and simplified Bishop method.

Outcomes:

Upon completion of this course, the students should be able to

- Understand the origin of the soil and geological cycle

- Apply principles of phase diagram for soil properties and perform basic weight-volume calculations
- Understand basics principle of flow and soil permeability through pores media including Bernoulli's equation Darcy's law and Hydraulic conductivity
- Calculate the insitu stress in saturated soil with and without seepage, seepage force, and implement measures to control heave in soil
- Estimate the amount of consolidation and settlement and time required for settlement under a given load
- Have knowledge of shear strength principle's including the Mohr's coulomb failure criteria

Suggested Readings:

1. Principals of Geotechnical Engineering, By: Braja M. Das., Fifth edition, First reprint 2002, low price edition, Thomson learning Inc.
2. Basic & Applied Soil Mechanics, By: GopalRanjan / Rao A.S.R. 2003 print, New Age International Pvt Ltd.
3. Lambe, T.W. And Whitman, R.V, "Soil Mechanics", John Wiley &Sones Inc., NY,1969.
4. Murthy, V.N. S. "Soil Mechanics & Foundation Engineering".DhanpatRai&Sons' 2006.
5. Arora, K.R., "Soil Mechanics and Foundation Engineering", Standard publishers Distributors, revised and enlarged sixth edition,2007.
6. Relevant IS Codes.

Category: **Professional Core Course**

Subject Code: **19CE3105**

DESIGN OF STEEL STRUCTURES

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To know the IS codal provisions as applicable for the designs.
- To understand the material behavior and basics of design of steel structures.
- To learn the design of various members along with the connections.
- To explain the design principles of roof trusses.

UNIT – I: (14hours)

Materials and specifications: Chemical composition of steel, types of structures steel, Residual stresses, stress Concentration.

Basis of Structural Design: Codes and specifications, Design Philosophies, Working stress method, Limit state method.

Loading and load Combinations: Characteristic loads, Dead loads, Imposed loads, Earthquake loads, and Wind loads and load Combinations. Partial safety factors for materials and loads.

Bolted Connections (Limit state method): Bolted Connections, Behavior of bolted joints Design Strength of ordinary Black bolts, Design strength of high strength friction grip bolts, pin connections and Eccentric connections.

Welded Connections (Limit state method): Advantages of Welding, types of welds and joints simple connections and Eccentric connections.

UNIT – II: (14hours)

Design of Tension Members (Limit state method): Types of tension members, design of strands, slenderness ratio, modes of failure, factors effecting strength of tension members, design of tension members (angles, other sections and rods), lug angles, tension member splice.

UNIT – III: (14hours)

Design of Compression members (Limit state method): introduction, possible failure modes. behavior of compression members, elastic buckling of slender compression members, behavior of compression members, sections of compression members, effective length design of compression members with single section and built-up sections (symmetric in both directions), lacing and battening column splices.

Design of column bases (Limit state method): design of slab base and gusseted base for columns.

UNIT – IV: (10hours)

Design of beams (Limit state method): Types of beams, section classification, lateral stability of beams, buckling of real beams, behavior of beams in bending, design of laterally supported and unsupported beams, design of compound beams, shear strength of beams, maximum deflection, web buckling crippling, biaxial bending and unsymmetrical bending.

UNIT – V: (8hours)

Design of Roof trusses (Limit state method): Types of trusses, end bearings, spacing of trusses and purlins, estimation of loads with different roof coverings, self -weight of truss, wind effects, design of purlins for dead load and wind loads. Detailed design of roof trusses including joints and supports (only angular trusses).

Outcomes:

Upon completion of this course, the students should be able to

- Appraise the importance of steel structures and limit state method of design
- Design different types of bolted and welded connections
- Classify and design various types of tension and compression members
- Design different flexural members and column bases

Suggested Reading:

1. Subramanian. N, design of steel structures, Oxford University Press, 2008.
2. Shiyekar M.R., Limit State Design in Structural Steel, PHI Learning Pvt.Ltd. 2010.
3. Bhavikatti, SS: Design of Steel Structures:, I.K International Publishing House Pvt.Ltd 2010
4. Duggal S.K. Design of Steel Structures, Tata McGraw Hill Publishing, 2009.

Category: **Professional Core Course**

Sub code: **CE3106**

WATER RESOURCES ENGINEERING - II

Pre Requisites: Water Resources Engineering -I

Objectives: To study various types of reservoirs and diversion headwork.

The subject also covers the topics such as canals, gravity dams and earth dams.

Course Outcomes:

At the end of the course, the student will be able to: Plan an Irrigation System

- Design irrigation canals and canal network
- Plan and design diversion head works
- Design irrigation canal structures, knowledge on CD works
- Analyze gravity and earth dams
- Design spillways and energy dissipations works
- Analyze and design gravity dams

UNIT - I Storage Works-Reservoirs - Types of reservoirs, selection of site for reservoir, zones of storage of a reservoir, reservoir yield, estimation of capacity of reservoir using mass curve Reservoir Sedimentation – Life of Reservoir. Types of dams, factors affecting selection of type of dam, factors governing selection of site for a dam.

UNIT - II Gravity dams: Forces acting on a gravity dam, causes of failure of a gravity dam, elementary profile, and practical profile of a gravity dam, limiting height of a low gravity dam, Factors of Safety - Stability Analysis.

Earth dams: types of Earth dams, causes of failure of earth dam, criteria for safe design of earth dam, seepage through earth dam, measures for control of seepage.

Spillways: types of spillways, Design principles of Ogee spillways - Spillway gates. Energy Dissipaters and Stilling Basins

UNIT- III Canal systems: Alignment, classification of alluvium canals and their functions, regime concept of Kennedy's and Lacey's theories, design of canals based on Kennedy's and Lacey's method. Use of Garrett's diagrams for the design of canals, lining of canals, methods of lining and design of lined canals.

Introduction of **cross drainage works** – types - selection of suitable type of CD works – aqueduct and siphon – super passages-level crossing-necessity.

UNIT- IV Diversion Head works: Types of Diversion head works- weirs and barrages, layout of diversion head work - components. Causes and failure of Weirs and Barrages on permeable foundations, -Silt Ejectors and Silt Excluders Weirs on Permeable Foundations – Creep Theories - Bligh's, Lane's and Khosla's theories, Determination of uplift pressure- Various Correction Factors – Design principles of weirs on permeable foundations using Creep theories - exit gradient, U/s and D/s Sheet Piles - Launching Apron.

UNIT- V Canal Falls - types of falls and their location, Design principles of Notch Fall and Sarada type Fall.

Canal regulation works, principles of design of distributor and head regulators, Canal Cross Regulators -canal outlets, types of canal modules, proportionality, sensitivity and flexibility.

Hydropower- classification, and principal components of hydroelectric power plants.

TEXT BOOKS:

1. Irrigation and water resources engineering by G.L. Asawa, New Age International Publishers.
2. Irrigation engineering by K. R. Arora Standard Publishers.
3. Irrigation and water power engineering by Punmia & Lal, Laxmi publications Pvt. Ltd., New Delhi

REFERENCES:

1. Theory and Design of Hydraulic structures by Varshney, Gupta & Gupta
2. Irrigation Engineering by R.K. Sharma and T.K. Sharma, S. Chand Publishers 2015.
3. Irrigation Theory and Practice by A. M. Micheal Vikas Publishing House 2015.

Category: **Professional Core Course**

Subject Code: **CE3802**

HIGHWAY ENGINEERING LAB

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-2-1

List of experiments:

A) Test on Bitumen

1. Penetration Test
2. Ductility Test
3. Softening point test
4. Flash
and fire
point test
5. Viscosity
test

B) Test on road aggregate

6. Aggregate crushing value test
7. Los Angeles abrasion test
8. Aggregate impact value test
9. Aggregate Particle Size Distribution & Shape tests (flakiness & elongation)
10. Specific Gravity
11. Water Absorption
12. Soundness (Demonstration only)

C) Tests on soil

13. Determination of CBR

Outcome:

Upon completion of this course,

- Student gains knowledge about testing the properties of aggregates, bitumen and soil with the help of various tests

Suggested Reading:

1. Relevant IS Codes of Practice
2. Relevant ASTM and AASHTO Codes of Practice
3. Khanna, S K and Justo C. E. G., Highway material Testing (laboratory manual). Nemchand and Bros , Roorke (2000).

Category: **Professional core course**

Subject Code: **CE3702**

SOIL MECHANICS LAB

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-2-1

List of experiments:

1. Determination of moisture content (water content) of soil.
2. Determination of the specific gravity of soil particles finer than 2 mm.
3. Grain size analysis – Mechanical Method
4. Grain size analysis – Hydrometer analysis
5. Atterberg limits determination – Liquid Limit, Plastic Limit, Shrinkage Limit
6. Determination of the in-situ density of soils by core cutter method or sand replacement method.
7. Determination of the max. dry density – optimum moisture content for a soil from Proctor test.
8. Determination of the Coefficient of permeability of the soil – Falling head method.
9. Determination of one-dimensional consolidation parameters of an undisturbed cohesive soil sample from consolidation test.
10. Determination of shear strength parameters of a silty or sandy soil at known density and moisture content by direct shear test.
11. Determination of shear strength parameters of the fine grained soil by Triaxial tests – UC,

UU, CU, CD tests.

Outcome:

Student know the technique to determine index property and engineering properties such as shear strength, compressibility and permeability by conducting appropriate tests

Technical Seminar-III

Category: **Professional Core Course**

Subject Code: **CE3201**

CONSTRUCTION TECHNOLOGY AND PROJECT MANAGEMENT

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To know the Fundamentals of construction practices and procedure of construction
- To know about various methods and equipment, safety practices generally adopted in construction
- To understand the importance of contract management and different techniques
- To introduce the students about project management, techniques used, cost analysis with respect to civil engineering projects.

UNIT-I: (8 hours)

Introduction to construction technology – Fundamentals- construction activities and schedules- construction records and documents- quality and safety- codes and regulations

Construction Equipment and Methods-factors effecting selection of construction equipment- technical and economic aspects-analysis of production outputs and costs-methods and equipment for earthmoving-lifting and erection-material transport –pile driving-dewatering-batching and mixing- transport and placement-finishing-formwork

UNIT-II: (6 hours)

Safety engineering – Principles on safety- personnel, fire and electrical safety-classification of construction accidents and causes- direct and indirect loss due to accidents- Local hazards and their elimination- safety in demolition of buildings- safety in storage and handling of materials and equipment

UNIT-III: (10 hours)

Project Management- Introduction to project management-Project planning-Scheduling and controlling-Objectives of planning-Stages in construction-Sequences of events in civil engineering projects-Construction schedule-Project management tools

UNIT-IV: (14 hours)

Project management and Techniques- Development of management techniques-bar charts-Gantt charts- CPM and PERT techniques-Problems on CPM and PERT – Network analysis-problems on Network analysis

UNIT-V: (12 hours)

Cost Analysis and Resources allocation- Introduction to cost analysis-cost reduction in construction management- cost time analysis- crashing the network-optimization –resource allocation –resource leveling and smoothing-introduction o project management software's-MS Project, Gantt charts etc.,

Outcomes:

At the end of the course students will be able to

- Understand the procedure to carry out the construction by selection of suitable method and equipment.
- Prepare the construction site with adopting suitable safety measures to avoid accidents
- Calculate the total cost, economy and time period needed for completion of project by contract management
- Evaluate the procedure of contract and tendering procedure in practice

Suggested Reading:

1. Construction Project Management -Kumar Neeraj Jha -Pearson Publication -2015
2. Puerifoy R.L. -Construction Planning Equipment & methods
3. Punmia and Khandelwal K.K. -Project Planning and Control -Laxmi Publ. Delhi.
4. Mahesh Varma -Construction Planning and Equipment -Metropolitan Co.
5. Choudhary S. -Project Management -Tata McGraw Hill Publishing Company Limited, New Delhi

ENVIRONMENTAL ENGINEERING

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To fill the gap between general introductory environmental science and the more advanced environmental engineering.
- To explain the different sequential unit operations of water and wastewater treatment processes.
- To provide necessary engineering principles for analyzing the environmental issues
- To motivate the present and future generations to take suitable care of sustainability of all existing resources

UNIT – I: (10 hours)

Water Supply and Natural Phenomenon: Objectives of protected water supplies; Rate of demand; Population forecasts; Surface and Groundwater sources of water; Water analysis; Self Purification of Streams.

UNIT – II: (12 hours)

Water treatment and Distribution: Classification, description and design of: Coagulation, Flocculation, sedimentation, filtration, disinfection and softening process; Methods of Layout of Distribution pipes; Design of Distribution by Hardy Cross method for simple networks.

UNIT – III: (10 hours)

Wastewater and Primary Treatment: Wastewater Characteristics; Standards of Disposal; Treatment Objective and, Strategies; Layouts of Primary, Secondary and Advanced Treatment Units; Design of Preliminary and Primary Treatment Operations: Screens, Grit Chambers, Skimming Tank, Primary and Secondary Sedimentation Tanks.

UNIT – IV: (10hours)

Biological Treatment: Classification and Design of: Activated Sludge Process; Oxygen and Nutrient Requirements; Principles of Trickling Filter Process, Oxidation Ponds, Lagoons.

UNIT V: (10 hours)

Sludge Treatment and Disposal: Sludge Thickening; Aerobic and Anaerobic Sludge Digestion Processes; Design of Digester Tank, Sludge Dewatering.

Outcomes:

Upon completion of this course, the students should be able to

- Assess characteristics of water and waste water and their impacts.
- Estimate the population of existing town and demand of water.
- Know the self purification of rivers thus standard of releasing the waste in to them.
- Estimate quantities of water and waste water and plan conveyance components.
- Design components of water and waste water treatment plants.

Suggested Reading:

- 1.Metcalf and Eddy, Wastewater Engineering - Collection, Treatment, Disposal and Reuse, McGraw Hill Pub. Co., 1995.
- 2.Nelson Leonard Nemerow, Industrial Waste Treatment, Butterworth-Heinemann, 2007.
- 3.Benefield L.D. and Randall C.D. Biological Process Designs for Wastewater Treatment, Prentice Hall Pub. Co., 1980.
- 4.P. N. Modi- Sewage Treatment Disposal and Waste Water Engineering.

Professional Elective-I

Professional Elective-II

ESSENCE OF INDIAN TRADITION AND KNOWLEDGE

Subject Code:

Externals: 60Marks

L-T-P-C

Internals: 40Marks

2-0-0-0

UNIT –I Basic Structure of Indian Knowledge System:

Veda –Definition – Kinds –Upavedas (Ayurveda, Gandhra veda, Shilpa veda, Artha veda)- Vedangas (Shiksha, Kalapa, Chhanda, Niruktha, Vyakarana, Jyothishya), Dharma Shastra, Mimansa, Purana, Tarka Shastra

UNIT – II Modern Science and Indian Knowledge System

Yoga Holistic Health Care

UNIT – III Indian Philosophical Tradition

A) Orthodox School: Samkya, Yoga, Nyaya, Vaisheshika, Purva Mimansa, Vedantha

B) Heterodox School: Jainism, Buddhism, Ajivika, Anjana, Charvaka

UNIT-IV

Indian Linguistic Tradition

UNIT –V Indian Artistic Tradition:

Chithra Kala (Painting), Sangeetha Kala (Music), Nruthya Kala (Dance)

Sub code: **BM0007**

MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS

Externals:

60Marks L-T-P-C

Internals:

40Marks 3-0-0-3

Course Objective:

- Enable the students to learn managerial economics principles applied in industries and equip them to handle the tasks in their career by making a real sense of what is happening economically in the organization.
- The course describes the Nature and Scope of Managerial Economics. It gives complete study on the demand and elasticity of demand and methods of demand forecasting.
- It provides a detailed structure on the pricing strategies and shows clear picture methods and sources of raising finance.
- It gives clear cut information of preparing final accounts and capital Budgeting techniques.

Course Outcome:

After the successful completion of this course, the learner will be able to know:

1. The dynamic game of demand and supply, and how the trinity of Economics i.e. Demand, Supply and Scarcity make the things move around the globe.
2. Principles of Microeconomics applied to industries.
3. Concept of forecasting and applying forecasting techniques to address the challenges and opportunities in the organization they work.
4. Cost and Production analysis, Break-Even analysis, Opportunity Cost, how to optimize organizational resources and how to minimize cost and maximize production, revenue and profit
5. Different pricing structure and discount mechanism suitable for business firms.
6. Market structure and how to exploit market structure for optimizing the benefits of organization.
7. Capital requirements and sources of capital.

UNIT I: Introduction to Managerial Economics:

Definition, Nature and Scope of Managerial Economics-Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.

UNIT II: Theory of Production and Cost Analysis:

Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs. Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts, Opportunity cost. Fixed vs. Variable costs, Explicit costs Vs. Implicit costs. Out of pocket costs vs. Imputed costs. Break-even Analysis (BEA)- R22, Electronics and Communications Engineering Page 81 Determination of Break-Even Point (simple problems)- Managerial Significance and limitations of BEA.

UNIT III: Markets & Pricing Policies:

Market structures: Types of competition, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Objectives and Policies of Pricing- Methods of Pricing: Cost Plus Pricing, Marginal Cost Pricing, Sealed Bid Pricing, Going Rate Pricing, Limit Pricing, Market Skimming Pricing, Penetration Pricing.

UNIT IV: Introduction to Financial Accounting:

Introduction to Financial Accounting: Double entry Book Keeping, Journal, Ledger, Trail Balance and Final Accounts (Trading account, Profit and Loss Account and Balance sheet with simple adjustments).

UNIT V: Capital and Capital Budgeting:

Capital and Capital Budgeting: Capital and its significance. Types of Capital. Estimation of Fixed and Working capital requirements. Methods and sources of raising finance. Nature and scope of capital budgeting, features of capital budgeting proposals. Methods of Capital Budgeting: Payback Method. Accounting Rate of Return (ARR) and Net Present Value Method, Internal Rate of Return (IRR).

Reference Books:

1. Aryasri: Managerial Economics and Financial Analysis, TMH, 2009.
2. Varshney & Maheswari : Managerial Economics, Sulthan Chand, 2009.
3. Raghunatha Reddy & Narasimhachary: Managerial Economics & Financial Analysis, Scitech. 2009.
4. V. Rajasekarn & R. Lalitha. Financial Accounting, Pearson Education. New Delhi. 2010
5. Suma Damodaran, Managerial Economics, Oxford University Press. 2009.

Category: **Humanities and Social Sciences including Management Courses**
Subject Code: **19HS3202**

EFFECTIVE TECHNICAL COMMUNICATION –II

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-2-1

Module 1: Managing Time; Personal memory, Rapid reading, taking notes; Complex problem solving; Creativity

Module 2: Communication and Technical Writing- Public speaking, Group discussion, Oral; presentation, Interviews, Graphic presentation, Presentation aids, Personality Development. Writing reports, project proposals, brochures, newsletters, technical articles, manuals, official notes, business letters, memos, progress reports, minutes of meetings, event report.

Module 3: Ethics- Business ethics, Etiquettes in social and office settings, Email etiquettes, Telephone Etiquettes, Engineering ethics, Managing time, Role and responsibility of engineer, Work culture in jobs, Personal memory, Rapid reading, Taking notes, Complex problem solving, Creativity.

Outcome:

The student will acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills

Text/Reference Books:

1. David F. Beer and David McMurrey, Guide to writing as an Engineer, John Willey. New York, 2004
2. Diane Hacker, Pocket Style Manual, Bedford Publication, New York, 2003. (ISBN 0312406843)
3. Shiv Khera, You Can Win, Macmillan Books, New York, 2003.
AICTE Model Curriculum for Undergraduate degree in Civil Engineering (Engineering & Technology)
4. Raman Sharma, Technical Communications, Oxford Publication, London, 2004.
5. Dale Jungk, Applied Writing for Technicians, McGraw Hill, New York, 2004.
(ISBN:07828357-4)
6. Sharma, R. and Mohan, K. Business Correspondence and Report Writing, TMH NewDelhi 2002.
7. Xebec, Presentation Book, TMH New Delhi, 2000. (ISBN 0402213)

COMPUTATIONAL ANALYSIS LAB

Externals: 60Marks

L-T-P-C

Internals: 40Marks

0-0-2-1

Objectives:

- Analysis and design of beams using STAAD PRO and validation by manual method
- Analysis and design of columns, footings and slabs using STAAD PRO
- Analysis and design of portal frame using STAAD PRO and validation by manual method
- Analysis of beams and slabs using ANSYS and validation by manual method

List of experiments:

1. Analysis and design of RCC beams using STAAD PRO and validation by manual method
2. Analysis and design of RCC columns using STAAD PRO and validation by manual method
3. Analysis and design of RCC slabs using STAAD PRO and validation by manual method.
4. Analysis and design of one storey RCC portal frame using STAAD PRO and validation by manual method
5. Analysis and design of two storey RCC portal frame using STAAD PRO and validation by manual method
6. Analysis and design of isolated RCC footings using STAAD PRO and validation by manual method
7. Analysis and design of combined RCC footings using STAAD PRO and validation by manual method
8. Analysis and design of steel beams using STAAD PRO and validation by manual method
9. Analysis and design of steel columns using STAAD PRO and validation by manual method
10. Analysis of plane and space truss
11. Analysis of plane and space frame
12. Analysis of beams using ANSYS and validation by manual method
13. Analysis of slabs using ANSYS and validation by manual method

Outcomes:

After completion of this laboratory course, students will be able to

- Analyze and design beams using STAAD PRO and validate by manual method
- Analyze and design columns, footings and slabs using STAAD PRO
- Analyze and design portal frame using STAAD PRO and validate by manual method
- Analyze beams and slabs using ANSYS and validate by manual method

Laboratory Manual:

1. “Civil Engineering Software Applications Laboratory Manual”, prepared by the faculty of Civil Engineering.

Textbooks:

1. Manual of STAAD.Pro V8i”, Bentley software.
2. “Manual of ANSYS”.
3. S. Unnikrishna Pillai and Devdas Menon, “Reinforced Concrete Design”, Tata McGraw-Hill Publishing Company Limited, New Delhi, 3rd edn., 2011
4. A.K. Jain, “Limit State Design”, Nem Chand Brothers, Roorkee, 7th edn. 2012.

Category: Professional core course

Subject Code: **CE3802**

ENVIRONMENTAL ENGINEERING LAB

Externals: 60 Marks

L-T-P-C

Internals: 40 Marks

0-0-2-1

Objectives:

- To perform the experiments to determine water and waste water quality
- To understand the water & waste water sampling, their quality standards
- To estimate quality of water, waste water, Industrial water

List of experiments:

1. To determine Electrical Conductivity of water sample.
2. To determine the pH value of water sample by pH Meter.
3. To determine the Alkalinity of water sample.
4. To determine the Hardness value of water sample.
5. To determine the Turbidity water sample by Nephelometric method.
6. To determine the Sodium content of water sample.
7. To determine the Chloride content of water sample.
8. To determine the Bromide content of water sample.
9. To determine the Fluoride content of water sample.
10. To determine the Nitrogen, Nitrate content of water sample.
11. Determination of COD.
12. To determine the DO value of given water sample by DO meter.

13. Determination of BOD

14. To determine the Faecal Coliforms.

15. Determination of Total Suspended Solids (TSS) and Total Dissolved Solids (TDS) in given water sample.

Outcomes:

After the completion of the course student should be able to

- Understand about the equipment used to conduct the test procedures
- Perform the experiments in the lab
- Examine and Estimate water, waste water, air and soil Quality
- Develop a report on the quality aspect of the environment

Theme Based Project

Technical Seminar-IV

Category: **Humanities and Social Sciences including Management Courses**

Subject Code:

HS4101

PROFESSIONAL PRACTICE, LAW & ETHICS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

2-0-0-2

Objective:

- To understand the Legal and Regulatory Framework for doing business in India.

UNIT - I:

Professional Practice – Respective roles of various stakeholders: Government and Standardization Bodies (ex. BIS, IRC) Professional bodies (ex. IEI, IRC, ECI, Local Bodies/ Planning Authorities), Clients/ owners, Developers, Consultants, Contractors, Service agencies

Professional Ethics – Definition of Ethics, Professional Ethics, Engineering Ethics, Personal Ethics; Code of Ethics as of Institution of Engineers; Professional Responsibility, Professional

Ethics; Conflict of Interest, Gift Vs Bribery, Environmental breaches, Negligence, Deficiencies in state-of-the-art

UNIT – II:

ETHICS IN FUNCTIONAL AREAS OF BUSINESS- Ethics in Marketing: Ethical practices in product packaging and labeling - Pricing - Advertising -Direct marketing – Green marketing - Ethical vs. Unethical marketing behavior. Ethics in HRM: Ethical implications of Privacy – Harassment – Discrimination – Whistle blowing. Ethics in Finance: Accountability – Window dressing and disclosure practices – Insider trading.

UNIT –III:

Law of Contract: Nature of Contract and Essential elements of valid contract, Offer and Acceptance, Consideration, Capacity to contract and Free Consent, Legality of Object, Performance and discharge of Contracts, Remedies for breach of contract.

Unit-IV

Contracts-II: Indemnity and guarantee, Contract of Agency, Sale of goods Act -1930: General Principles, Conditions & Warranties, Performance of Contract of Sale.

Unit- V:

Law relating to Intellectual Property:Introduction – meaning of intellectual property, main forms of IP, Copyright, Trademarks, Patents and Designs, Secrets;

Outcome:

Students will be able to understand a) Professional &Business Laws related to contracts b) Importance of Ethics in Business c) IPR and Legal Aspects.

Suggested readings:

1. Ethics in Engineering- M.W.Martin& R.Schinzinger, McGraw-Hill
2. Maheshwari & Maheswari - A Manual of Business Laws, Himalaya Publishing House.
3. D. Chandra Bose - Business Law PHI-Private Limited, New Delhi.
4. A.C. Fernando - Business Ethics An Indian Perspective Pearson Education
5. Manuel G. Velasquez - Business Ethics Concepts and Cases Prentice-Hall of India Pvt.Ltd, 2008.

REMOTE SENSING AND GIS

Scheme of Evaluation:

External – 60 marks (Course – 40 marks, Lab – 20 marks) **L-T-P-C**

Internal – 40 marks (Course – 30 marks, Lab – 10 marks)

3-0-2-4

Objectives:

- Understands the basic concept of Remote Sensing and know about different types of satellite and sensors.
- Illustrate Energy interactions with atmosphere and with earth surface features.
- Knows satellite image processing and satellite image enhancement techniques.
- Can interpret satellite images, top sheet maps and integrating the data with GPS.
- Understand different components of GIS and learning about map projection and coordinate system
- Develop knowledge on conversion of data from analogue to digital and working with GIS software.
- Applying principles of Remote Sensing and GIS to collect, map, retrieve spatial information to plan, assess and evaluate natural and manmade systems using geospatial models and methods for hazard mitigation and resource planning.
- Can Pursue research and develop capabilities to handle multi-disciplinary field projects.

UNIT 1: (8 hours)

Introduction: Satellite based remote sensing, advantages, platforms of remote sensing, electromagnetic spectrum, interaction of EM spectrum with atmosphere, scattering, absorption and emission, Multi-spectral scanners and devices, Characteristics of different satellites (Landsat, IRS, Cartosat and Resourcesat), Limitations of remote sensing.

UNIT II: (10 hours)

Digital Image processing: Image characteristics and resolutions, Georeferencing, Principles of Image enhancement, spatial filtering, Band ratioing, Principal component analysis, Color representations and transformations, Limitations of Image processing.

UNIT IV: (10 hours)

Digital Image interpretation: Identification of different geological features, Image classification, Image histogram and statistics, Image merging and mosaicing, Image processing using software, using MATLAB, Applications of Image analysis, Remote sensing integration with GPS.

UNIT IV: (12 hours)

Geographic Information System: Introduction, History of GIS, Components of GIS, Maps and scale, Coordinate system, Map projection, Data storage, Spatial database management, Data models and structures (Vector and Raster), Geodatabase and Metadata, Integration of Remote Sensing with GIS, Web GIS, Limitations of GIS.

UNIT V: (12 hours)

GIS Analysis: Introduction to ARC GIS software, Georeferencing and transformations, Spatial data editing, Spatial analysis, Non spatial analysis, Cartographic principles and design, Map layout, Map designing, Spatial Interpolation techniques, Buffering, Spatial Querying, Topology.

Applications: Hands on training by performing Digital elevation modeling, Overlay operations, Image Classification, Land use and Land cover mapping, Watershed delineation.

Outcomes:

- Upon completion of course, students will be able to develop analytical workflows to derive products and extract information from remotely sensed data for a broad range of applications.
- Can process remotely sensed data to make it useful in geographic information systems.
- Will be able to extract information from remotely sensed data using a variety of manual and automated techniques;
- Designs multi-criteria geospatial systems for decision making process.
- Develops multi-step remote sensing workflows to solve problems in a variety of application areas;

Suggested Reading:

- Remote Sensing and Image Interpretation, Lillesand T.M & Kiefer R.W., John Wiley and Sons, 2008.
- Rees, W. G., Physical principles of Remote Sensing, Cambridge University Press, 2001.
- Remote Sensing and GIS B. Bhatta by Oxford Publishers 2015.
- Introduction to Geographic Information System – Kang-Tsung Chang, McGraw-Hill 2015
- Paul Longley., Geographic Information systems and Science, John Wiley & Sons, 2005.
- C.P.Lo & Albert K. W.Yeung, Concepts and Techniques of Geographic Information Systems, Prentice Hall India Pvt.Ltd, 2002.

List of Professional Electives

Sl. No	Code	Subject	Semester	L-T-P	Credits
1.	CE3211	Foundation Engineering	VI	3-0-0	3
2	CE3212	Railway and Airport Engineering	VI	3-0-0	3
3	CE3213	Finite Element Methods	VI	3-0-0	3
4	CE3214	Ground Water Modelling	VI	3-0-0	3
9	CE4115	Solid and Hazardous Waste Management	VII	3-0-0	3
10	CE4116	Sustainable Construction Management	VII	3-0-0	3
11	CE4117	Ground Improvement Techniques	VII	3-0-0	3
12	CE4118	Transportation Economics	VII	3-0-0	3
13	CE4119	Prestressed Concrete	VII	3-0-0	3
14	CE4120	Advanced Fluid Mechanics	VII	3-0-0	3
15	CE4121	Rehabilitation & Retrofitting of Structures	VII	3-0-0	3
17	CE4212	Earthquake Engineering	VIII	3-0-0	3
18	CE4213	Environmental Impact Assessment	VIII	3-0-0	3
19	CE4214	Construction Equipment & Automation	VIII	3-0-0	3
20	CE4215	Pavement Construction and Management	VIII	3-0-0	3

FOUNDATION ENGINEERING

Externals: 60Marks

Internals: 40Marks

L-T-P-C

3-0-0-3

Objectives:

- To enable application of soil mechanics principles learned in previous semester to the practice of foundation engineering
- To enable computation of stress distribution in soils due to applied loads
- To understand suitability of shallow/deep foundations and their Geotechnical design
- To learn about different methods of geotechnical investigations and its role in selection and design of foundation

UNIT - I

Earth Pressures: States of earth pressure – Active, passive, at rest condition; Rankine's theory computation of active and passive earth pressure in cohesion less and cohesive soils, smooth and rough walls, inclined backfill, depth of tension crack, Earth retaining structures – gravity, cantilever, counter fort, reinforced earth, etc.

UNIT - II

Geotechnical Investigations: Necessity -Principles of exploration -objectives-Soil profile-collection of distributed & undistributed soil samples-samplers & quality of samples, In-situ tests -SPT, CPT, plate load test.

Stresses distribution in soils: Boussinesq's theory-Computation of increment in vertical stress due to application of a point load (its distribution on horizontal, vertical planes), uniformly distributed circular and rectangular areas -Pressure bulb -Significant depth Construction and use of Newmark's chart -Westergaard's theory -Validity of elastic theories -Contact pressure distribution.

UNIT - III

Introduction to foundations: Functional requirements- types- shallow and deep foundations-suitability

Shallow Foundations: Types of shear failures, Terzaghi's theory for safe bearing capacity of shallow foundations -Effect of type of shear failure/shape of the footing /water table, ultimate and allowable bearing capacity, Settlements of foundations- immediate, consolidation and creep settlements. Types of shallow Foundations-Isolated, combined, mat/raft foundations etc.

UNIT - IV

Pile Foundation: Piles, pile groups, Types of pile foundations based on load transfer mechanism, material type, mode of installation, functional use. Shaft and base resistances, down drag, negative skin friction, pile load tests, Estimation of vertical load carrying capacity of a single pile -static formulae/Dynamic formulae/Pile load tests-Cyclic pile load tests for separation of total capacity into bearing and friction components-Pile groups-necessity -efficiency of Pile groups-estimation of group capacity-Concept of Piled raft foundation.

UNIT - V

Cassions: Necessity -Types-Essential components of open(well)/ box(floating)/ Pneumatic cassions-Suitability-Sinking of caissons-correction for tilt &shift, dewatering.

Coffer dams: Necessity-types-suitability

Underpinning: Necessity-methods (pin/pile)-suitability

Geo-synthetics: classification-functions-applications.

Outcomes:

Upon completion of this course, the students should be able to

1. Understand how stress are transferred through soil and be able to compute both geostatic and induced stress due to point line and the area load
2. Basic understand of lateral earth pressure concept and theory including Rankine's theory of active and passive earth pressure with and without sloping backfill
3. Understand the basic concept of ultimate bearing capacity of shallow foundations including modification of bearing capacity equation of water table, factor of safety and eccentrically loaded foundations

Suggested Readings:

1. Principals of Geotechnical Engineering, By: Braja M. Das., Fifth edition, First reprint 2002, low price edition, Thomson learning Inc.
2. Foundation Analysis and Design, By: Joseph E. Bowles., Fifth Edition, 2001 print, The McGraw Hill Companies Inc.
3. Basic & Applied Soil Mechanics, By: GopalRanjan / Rao A.S.R. 2003 print, New Age International Pvt Ltd.
4. Arora, K.R., "Soil Mechanics and Foundation Engineering", Standard publishers Distributors, revised and enlarged sixth edition,2007.

RAILWAY AND AIRPORT ENGINEERING

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To understand the geometric designing concepts in airport and railway engineering
- This course imparts the student's knowledge of planning, design, construction, and maintenance of railway tracks.
- The student develops skills on airport planning and design with the prime focus on runway and taxiway geometrics.

UNIT – I (12)

Railway Planning and Design: Introduction to Railways, permanent way component parts and its functions. Rails - various types, functions, creep in rails, creep measurement, coning of wheels and rail fixations. Sleepers - various types. Merits and demerits, ballast, various types and sub grade preparation. Railway alignment and geometric design - alignment. super-elevation, negative super elevation, cant deficiency. example problems.

UNIT – II (10)

Railway Track Construction, Maintenance and Operation: Points and crossing. layout of left and right hand turnouts. Signaling, Interlocking & Track Circuiting Construction and maintenance of permanent way.

UNIT –III (10)

Airport Planning: Introduction to air transportation, history and international organizations role in development of airports, air craft types and its characteristics. General lay-out of an airport and its component parts. Site selection of an airport as per ICAO

UNIT –IV (10)

Airport Design: Runway design, orientation of runway by wind rose diagrams,

basic runwaylength determination, corrections to basic runway length, geometric design, elements oftaxiway design, types of airports as per landing & take-off and dimensions, configuration & pavement design principles.

UNIT –V (10)

Airport Layouts, Visual Aids and Air Traffic Control: Apron, Terminal building, hangars, motor vehicle parking area and circulation pattern, principles of passenger flow, passenger facilities, visual aids, runway and taxiway markings, wind direction indicators, air traffic control, basic actions.

Outcomes:

- Gain Knowledge related to geometric design, operation and control of railways
- Gain Knowledge about different types of airports and airport runway design

References:

1. Khanna. S. K. .Arora, M. G. and Jain. S. S. (1994) "Airport Planning and Design" Fifth edition. Nem Chand & Bros, Roorkee, India.
2. Chandra, S and Agarwal, M. M. (2007) "Railway Engineering" Oxford Higher Education, University Press New Delhi.
3. Principles of Traffic Engineering – Garber & Hoel, Cengage Learning.
4. L.R. Kadiyali, *Principles and Practice of Highway Engineering*, Khanna Publications, New Delhi, 2000
5. Haas and Hudson W. R., "Pavement Management Systems" McGraw Hill publications.

FINITE ELEMENT ANALYSIS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To read Concept & Solution procedure for finite element displacement approach.
- To analyze Lagrangian & Hermitian interpolation functions.
- To illustrate shape function & numerical integration technique.
- To understand element properties for one dimensional (bar & beam) element & two dimensional (rectangular, triangular & isoparametric elements using natural & area coordinate system.
- Introduction to plate elements, shell elements, dynamics and vibration, buckling, Galerkin Method

- Develop pre and post processors, Solution Techniques & Software Packages.

UNIT – I: (12 hours)

Introduction to Finite Element Method –Background and general description of the method – summary of the analysis procedure–Mathematical preliminaries-Linear function spaces, operators and functionals. Continuity and differentiability. Galerkin methods, Collocation methods, Least-squares methods. Variational methods of approximation.

Basic Equations in Elasticity – stress strain equations – concept of plane stress – plane strain— advantages and disadvantages of FEM. Element shapes – nodes – nodal degree of freedom – strain displacement relations – energy principles – Discretization – Rayleigh –Ritz method of functional approximation - variational theorems.

UNIT – II: (12 hours)

Theory of Finite element method – Concept of element – various element shapes – displacement models –Finite Element Analysis (FEA) of – one dimensional problems – Stiffness matrix for Bar element – Shape functions- stiffness matrix

FEA Beam elements – stress strain relation - Assemblage of elements construction of stiffness matrix and loads – boundary conditions and solution of overall problem – Application to continuous beams.

Principles of Elasticity:Equilibrium equations – straindisplacement relationships in matrix form – Constitutive relationships for plane stress, plane strain and Axi-symmetric bodies of revolution with axi-symmetric loading.

UNIT – III: (10 hours)

Two dimensional elements: FEA Two dimensional problem – CST – LST element – Polynomial shape function – stress – strain-Displacement models –generalized coordinates – Lagrangian – Serendipity elements – Hermite polynomials – regular, Irregular 2 D & 3D –Element –shape functions – convergent and compatibilityrequirements – Geometric invariance – Natural coordinate system – area and volume coordinates

UNIT – IV: (8 hours)

Isoparametric formulation:The concept of isoparametric mapping - Concepts of, isoparametric elements for 2D analysis –formulation of CST element, 4 –Noded and 8-noded iso-parametric quadrilateral elements –Lagrangian and Serendipity elements - Computer implementation - Application to elliptic parabolic and hyperbolic differential equations.

Generation of element: Generation of element stiffness and Nodal load matrices for 3-node triangular element and four noded rectangular elements.

UNIT-V:(8hours)

Axi-symmetric analysis: Basic Principles-Formulation of 4-noded iso-parametric axi-symmetric element

Solution techniques: Numerical Integration, Staticcondensation, assembly of elements and solution techniques for staticloads.

Outcomes:

Upon completion of course, students will be able to

- Develop shape functions and stiffness matrices for spring and bar elements Draw and detail different Components of different types of doors and windows.
- Develop global stiffness matrices and global load vectors,
- Prepare structural drawings with details of Beam, Slab and Footing.
- Apply natural and arial coordinate systems to constant strain triangle and linear strain triangle elements
- Analyze planar structural systems using finite element modeling

Suggested reading:

1. Introduction to Finite elements in engineering by Chandrupatla, Belegundu, Prentice Hall.
2. Finite element method by Daryl L. Logan, CENGAGE Learning.
3. Finite Elements Methods in Engineering by Tirupati. R. Chandrnpatla and Ashok D. Belegundu – Pearson Education Publications.
4. Finite element analysis by S.S. Bhavakatti-New age international publishers
5. Finite Element methods for Engineers by U.S.Dixit, Cengage Publishers, New Delhi.
6. Finite element analysis by P. Seshu, PHI.
7. Concepts and Applications of Finite Element Analysis by Robert D.Cook, David S. Malkus and Michael E. Plesha. John Wiley & Sons.
8. Finite element analysis by David V Hutton, Tata Mcgraw Hill, New Delhi
9. Applied Fem by Rammurthy, I.K.International Publishers PVt.Ltd.,New Delhi.
10. Fem by J.N.Reddy, Mcjraw, TMH Publications, New Delhi.

GROUND WATER MODELLING

Externals: 60Marks

Internals: 40Marks

L-T-P-C

3-0-0-3

Objectives:

- A pathway to understand the basic physical principles of groundwater flow, differential equations, boundary condition and groundwater quality.

- To gain knowledge of various aspects of recharge of groundwater.
- Exposure to use the numerical solutions to solve problems with complex realistic situations.

UNIT – I: (10 hours)

Ground Water Occurrence: Ground water hydrologic cycle, origin of ground water, rock properties effecting ground water, vertical distribution of ground water, zone of aeration and zone of saturation, geologic formation as Aquifers, types of aquifers, porosity, Specific yield and Specific retention.

UNIT – II: (12 hours)

Ground Water Movement: Permeability, Darcy's law, storage coefficient. Transmissivity, differential equation governing ground water flow in three dimensions derivation, ground water flow equation in polar coordinate system. Ground water flow contours their applications.

UNIT – III: (10 hours)

Steady groundwater flow towards a well in confined and unconfined aquifers – Dupuit's and Theim's equations, Assumptions, Formation constants, yield of an open well interface and well tests – Recuperation Test.

UNIT – IV: (8 hours)

Unsteady flow towards a well – Non equilibrium equations – Theis' solution – Jacob and Chow's simplifications, Leaky aquifers – Well Interference.

UNIT – V (10 hours)

Surface and Subsurface Investigation: Surface methods of exploration – Electrical resistivity and Seismic refraction methods. Subsurface methods – Geophysical logging and resistivity logging. Aerial Photogrammetry applications along with Case Studies in Subsurface Investigation.

Course Outcomes:

Upon completion of course, students will be able to

- Know about groundwater hydrology and hydraulics of the movement of water in aquifers to manage groundwater resources.
- Deal with more realistic situations to solve problems pertaining to groundwater quality.
- Conduct simulation studies for future state of groundwater systems, optimal protection and rehabilitation strategies

TEXT BOOKS:

1. Ground water Hydrology by David Keith Todd, John Wiley & Son, New York.
2. Groundwater by H.M.Raghunath, Wiley Eastern Ltd.

3. Groundwater Hydrology by BOWER, John Wiley & sons.
4. Groundwater System Planning & Management – R Willes & W.W.G.Yeh, Printice Hall.
5. Applied Hydrogeology by C.W.Fetta, CBS Publishers & Distributers.

SOLID AND HAZARDOUS WASTE MANAGEMENT

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Pre Requisite: Environmental Engineering

Objectives:

- To know about different types of wastes prevailing.
- To gain the knowledge of sources of generation of MSW and Hazardous waste.
- To know the basics of Radioactive waste characteristics and its causes.
- To impart the knowledge of various rules and regulations of handling the S&HW.
- To understand the different managing techniques & treatment methods of S&HW.
- To design the landfill for S&HW.

UNIT I: (4 hours)

Regulations: Municipal solid waste (Management and handling) rules; Hazardous waste (Management and handling) rules; Biomedical waste handling rules; flyash, recycled plastics usage, batteries usage - management and handling rules.

UNIT II: (10 hours)

Municipal Solid Waste Management: Sources; composition; generation rates; collection of waste; separation, transfer and transport of waste; treatment and disposal options.

Hazardous Waste Management: Characterization of waste; compatibility and flammability of chemicals; fate and transport of chemicals; health effects

UNIT III: (11 hours)

Radioactive Waste Management: Sources, measures and health effects; nuclear power plants and fuel production; waste generation from nuclear power plants; disposal options.

Environmental Risk Assessment: Defining risk and environmental risk; methods of risk assessment; case studies.

UNIT IV: (8 hours)

Physicochemical Treatment of Solid and Hazardous Waste: Chemical treatment processes for MSW - combustion, stabilization and solidification of hazardous wastes; physicochemical

processes for hazardous wastes - soil vapor extraction, air stripping, chemical oxidation; ground water contamination and remediation

UNIT V: (10 hours)

Biological Treatment of Solid and Hazardous Waste: Composting; bioreactors; anaerobic decomposition of solid waste; principles of biodegradation of toxic waste; inhibition; co-metabolism; oxidative and reductive processes; slurry phase bioreactor; in-situ remediation

Landfill design: Landfill design for solid and hazardous wastes; leachate collection and removal; land fill covers; incineration.

Outcomes:

Upon completion of course, students will be able to

- Know various rules and regulations of handling the wastes from sources to treatment units.
- Understand the treatment methods for S&HW and their disposal techniques.
- Gain the knowledge of Environmental risk and its assessment, suitable design

Text/Reference Books:

1. John Pichtel Waste Management Practices CRC Press, Taylor and Francis Group 2005.
2. LaGrega, M.D. Buckingham, P.L. and Evans, J.C. Hazardous Waste Management, McGraw Hill International Editions, New York, 1994.
3. Richard J. Watts, Hazardous Wastes - Sources, Pathways, Receptors John Wiley and Sons, New York, 1997.
4. Basics of Solid and Hazardous Waste Mgmt. Tech. by Kanti L. Shah 1999, Prentice Hall..

SUSTAINABLE CONSTRUCTION MANAGEMENT

Externals: 60 Marks

L-T-P-C

Internals: 40 Marks

3-0-0-3

Objectives:

- To learn the concept of green building and energy conservative techniques.
- To study various sustainable building materials and cost effective techniques for sustainable building
- To know the various environmental techniques for waste water management.

UNIT-I: (13 Lectures)

Concept of Green Building: Sustainable Development concept, Buildings and climate, important considerations for the design of sustainable buildings, Green Building Assessment, Current version of the LEED rating system.

Energy Conservation: The design of a sustainable building, Lighting - day lighting; Ventilation - natural ventilation; Indoor air quality; Passive and Active systems for energy production and conservation, Elements of successful design of a building envelope.

UNIT-II: (7 Lectures)

Sustainable Building Materials: Environmental issues related to building materials, Local, Building materials from agricultural waste and Industrial waste.

UNIT-III: (8 Lectures)

Cost Effective Techniques for Sustainable Building: Stabilized Mud blocks, Stone masonry blocks, solid and hollow concrete blocks, Selection of building blocks, Ferro- Concrete, Properties and Uses, Practical aspects.

UNIT-IV: (8 Lectures)

Alternative sustainable Roofing Systems: Concepts in Roofing alternatives, Filler slab roofs, Composite Slab panel roofs, hollow block roofs, Masonry Domes.

UNIT-V: (8 Lectures)

Environmental Techniques: Waste water Management, Rain water harvesting and conservation, recycling, waste water treatment processes, external drainage system in building.

Outcomes: At the end of the course, the students will be able to

1. Know various components which makes the building energy efficient such as lighting, space conditioning, heat control and energy efficient.
2. Know Cost Effective Techniques for Sustainable Building
3. Decide which are most sustainable building materials and which method can be opted for sustainable development
4. Know the environmental techniques for waste water management and drainage system in building

Suggested References:

1. K. S. Jagadish, B. V. V. Reddy, "Alternative Building Materials and Technologies", New Age International Publishers
2. Gevorkian "Green Buildings" Mac Graw hill.
3. P. N. Balaguru and S.P. Shah, "Fibre reinforced Cement Composites", McGraw Hill.
4. The engineering guide to LEED- new construction-sustainable construction for engineer's haselbach.
5. D. J. Hannant, John Wiley and Sons, Fibre cements and Fibre Concretes.
6. A. M. Neville, Properties of Concrete, ELBS, Longman.

7. Miller G. T Jr Living in the environment, Cengage Publisher.
8. Cunningham W, Principles of Environmental Science, TM

GROUND IMPROVEMENT TECHNIQUES

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Prerequisites: Soil Mechanics and Foundation Engineering

Objectives:

- To know the need of ground improvement
- To acquire the knowledge on the various ground improvement techniques available and their applications for different types of soils
- To understand suitable ground improvement technique for given soil conditions.

UNIT-I: (4 Hours)

Introduction: Need for Ground Improvement, Different types of problematic soils, Emerging trends in ground Improvement. Types of Stabilization

UNIT-II: (8 Hours)

Mechanical stabilization: Shallow and deep compaction requirements, Principles and methods of soil compaction, Shallow compaction and methods, Properties of compacted soil and compaction control, Deep compaction and Vibratory methods, Dynamic compaction.

UNIT-III: (9 Hours)

Hydraulic modification: Ground Improvement by drainage, Dewatering methods, Design of dewatering systems, Preloading, Vertical drains, vacuum consolidation, Electro-kinetic dewatering, design and construction methods.

UNIT-IV: (9 Hours)

Modification by admixtures: Cement stabilization and cement columns, Lime stabilization and lime columns, Stabilization using bitumen and emulsions, Stabilization using industrial wastes Construction techniques and applications.

UNIT-V: (10 Hours)

Grouting: Permeation grouting, compaction grouting, jet grouting, different varieties of grout materials, grouting under difficult conditions.

In situ soil treatment methods: Soil nailing, rock anchoring, micro-piles.

Case studies of ground improvement projects.

Outcomes:

At the end of the course, student will be able to

- Know the necessity of ground improvement
- Understand the various ground improvement techniques available
- Select & design suitable ground improvement technique for existing soil conditions in the field.

REFERENCES

1. Manfred R. Hausmann, Engineering Principles of Ground Modification, McGraw-Hill Pub, Co., 1990.
2. M C. R. Davies, F.Schlosser Ground improvement geosystems.
3. Koerner, R. M., Designing with geosynthetics, Prentice Hall Inc. 1998.

TRANSPORTATION ECONOMICS

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To be aware of the organizational structure of transport corporations and their interactions.
- To understand economic analysis of transport projects.

UNIT-I (10hours)

Introductory Concepts in Transportation Decision Making: Overall transportation project development, budgeting, financial planning, the process of transportation project development, models associated with transportation impact evaluation;

UNIT-II (8hours)

Transportation costs -Classification of transportation costs, transportation agency costs, transportation user costs, general structure and behavior of cost functions and road pricing.

UNIT-III (10hours)

Transportation costs - Supply and demand - elasticity of demand; Supply of transport services - Economics of traffic congestion - Pricing policy. Vehicle operating costs: Fuel costs - Maintenance and spares, Depreciation - Crew costs - Value of travel time savings - Accident costs.

UNIT-IV (12hours)

Economic analysis of projects - Methods of evaluation - Cost-benefit ratio, first year rate of return, net present value, and internal-rate of return methods; Indirect costs and benefits of transport projects.

UNIT-V (10hours)

Financing of road projects - methods – Private Public Partnership (PPP) - Toll collection - Economic viability of Design-Build-Operate-Transfer Schemes – Risk Analysis – Value for Money analysis - Case Studies.

Outcomes:

Upon completion of this course, the students should be able to

- Understand the Motor Vehicle Act and statutory provision for road transport. Also the students will be able to do scheduling of route, vehicle and crew.
- Design the depots and terminals
- Do economic evaluation of the transportation projects and will understand the concepts of the Private Public Partnership

Suggested Reading:

1. Winfrey, Economic analysis for Highways, International Textbook Company, Pennsylvania, 1969.
2. CRRI, Road User Cost Study in India, New Delhi, 1982
3. IRC, Manual on Economic Evaluation of Highway Projects in India, SP30, 2007

ADVANCED FLUID MECHANICS

Externals: 60Marks

Internals: 40Marks

L-T-P-C

3-0-0-3

Objectives:

- Introduction to advanced concepts of fluid motion
- Knowledge about boundary layer concepts
- Description of ideal fluid flow theory and its application in real life problems.

UNIT-I: (10 hours)

Basic concepts: One, two and three dimensional flows, Lagrangian and Eulerian description of fluid flows, acceleration of fluid particles, general motion of fluid element (Translation, Dialation, Rotation, and Angular deformation), general stress-strain system, equations of motion, continuity equation, velocity potential, stream function, rotational and irrotational flows, circulation and velocity.

UNIT-II: (8 hours)

Dynamics of flow: Navier-Stokes equations, simple exact solutions to NS equations (steady laminar flow between parallel plates and in a circular tube of constant diameter).

UNIT-III: (10 hours)

Boundary layer flows: Concept of boundary layer, Prandtl's boundary layer equations, Von Karman –Pohlhausen integral momentum equation, Blasius solution for laminar boundary layer flow over a flat plate, boundary layer separation.

UNIT-IV: (10 hours)

Turbulent Boundary layer: Source of turbulence, velocities, energies and continuity in turbulence, turbulent shear stress, Reynolds equations for incompressible fluids, Prandtl's mixing length theory in shear flows, law of wall, velocity defect law, velocity distribution in turbulent flow through smooth and rough pipes.

UNIT-V: (8 hours)

Standard patterns of flow: Source, sink, vortex pair, spiral vortex, flow past a half body, flow past a cylinder with and without circulation, flow past a Rankine body.

Outcomes:

Upon completion of course, students will be able to

- Comprehend the advanced concepts of fluid motion
- Apply boundary layer and ideal fluid flow theory concepts in water resources engineering problems

Suggested Reading:

1. Schlichting Herman and Klaus Gersten. 'Boundary Layer Theory', Springer(India) Pvt Ltd., New Delhi, 2007
2. Yunus A Cengel and John M Cimbala, 'Fluid mechanics: Fundamentals and Applications', Tata McGraw-Hill Publishing Company, New Delhi, 2008
3. Som, S.K. and Biswas, G., 'Fluid Mechanics and Fluid machines', Tata McGraw-Hill Publishing company, New Delhi 1998
4. Vijay Gupta and Santhosh K. Gupta, 'Fluid mechanics and its applications', Wiley Eastern Ltd., New Delhi, 1984.

REHABILITATION & RETROFITTING OF STRUCTURES

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To learn the fundamentals of maintenance and repair strategies.
- To study the quality assurance, serviceability and durability of concrete.
- To know the various materials and techniques used for repair of structures.
- To educate the different repair, strengthening, rehabilitation and retrofitting techniques.

UNIT-I: (10 Hours)

Durability of concrete: Factors affecting durability of concrete, Corrosion of reinforcements in concrete, Carbonation, Chloride ingress, Alkali-silica reaction, Freeze-thaw effects, Chemical attack, Abrasion, erosion and cavitation, Weathering and efflorescence

UNIT-II: (9 Hours)

Defects and deterioration in buildings, Survey and assessment of structural conditions in RCC structures. Damage/condition assessment and various methods (for quantification) for its evaluation, Rapid Visual Screening (RVS) and ways to do RVS of damaged/deteriorated structures, Overview of health monitoring techniques.

UNIT-III: (13 Hours)

Non-destructive testing (NDT) of concrete quality, Non-destructive testing of connections in steel, Corrosion assessment in reinforcements in RCC elements and components in steel structures; Design principles, techniques and working mechanism, various instruments used for NDT evaluations (for strength, durability etc.) like Rebound Hammer, UPV, impact echo etc.; Technology used in various advanced instruments like Imaging techniques, GPR, Thermography, Tomography etc.

UNIT-IV: (9 Hours)

Materials for repairs, rehabilitation and retrofitting processes, Methods for

repairs, rehabilitation and retrofitting including surface preparation, Study of failures of buildings and lessons learnt, Role of quality control in construction as Preventive Measures Maintenance of buildings.

UNIT-V: (4 Hours)

Strengthening of Earthquake-damaged buildings; Introduction to Push-over analysis.

Outcomes: At the end of the course, the students will be able to

1. Detect the defects and deterioration in buildings
2. Survey and assess structural conditions in RCC structures
3. Understand and apply rehabilitation and retrofitting processes
4. Understand non-destructive testing and interpretation of the results for concrete and steel structures.

Suggested References:

1. P.K. Mehta and P.J.M. Monteiro, "Concrete - Microstructure, Properties and Materials", McGraw-Hill, New York, 2014.
2. Barry A. Richardson, "Defects and Deterioration in Buildings", E & FN Spon Press, London, 1991
3. J. H. Bungey, "Testing of Concrete in Structures", Chapman and Hall, New York, 1989.
4. A.R. Santakumar, "Concrete Technology", Oxford University Press, New Delhi, 2006.
5. B.L. Gupta and Amit Gupta, 'Maintenance and Repair of Civil Structures', Standard Publications, New Delhi, 2010.
6. Peter H. Emmons, "Concrete Repair and Maintenance Illustrated", RS Means, John Wiley & Sons, New York, 1981.
7. W.H. Ransom, "Building Failures: Diagnosis and Avoidance", E & FN Spon Press, London, 1992.

EARTHQUAKE ENGINEERING

Externals: 60 Marks

Internals: 40 Marks

L-T-P-C

3-0-0-3

Objectives:

This course will develop student's knowledge on

- Basics of Earthquake engineering and terminology involved
- Analysis of Single and Multi Degree of Freedom systems
- Estimation of Storey shear and Torsional moments & seismic design of footings

UNIT I

Fundamentals of earthquakes: Geology of earth, configuration of tectonic plates in a globe, influence of Geology on earthquake, behavior of plates, their motion and effects, causes of earthquake and their characteristics, Earthquake parameters, magnitudes, intensity, scales, seismic zoning of India, seismic coefficients for different zones, Liquefaction, causes and its remedial measures.

Disaster Mitigation: Natural disasters, mitigation and social aspects. Lessons from past earthquake: -Study of damages caused due to past, earthquakes in/ outside India and remedial measures.

UNIT II

Introduction to Dynamic Loads: Static Load v/s Dynamic Load, Types of Dynamic forces, Force Control and Displacement Control

Single Degree of Freedom Systems: Vibrations, Definition, Causes, and classifications. Single Degree of Freedom systems (SDOF), free, forced, damped, un-damped vibrations. **Multi Degree of Freedom systems:** Introduction to Multi-degrees of Freedom systems (MDOF), Derivations of related equations and solutions to two degree of freedom system

UNIT III

Seismic design of Symmetrical RC structures: Effect of earthquake on RC structure, IS provision, Seismic coefficient method. Basic requirements, Estimation of story shear.

Seismic design of Unsymmetrical RC structures: Effect of unsymmetrical geometry and masses, mass center and stiffness center, Estimation of story shear and torsional moments for unsymmetrical buildings, IS 1893 provisions to response spectrum, Concept of ductile detailing, IS 13920 provisions for RC frame

UNIT IV

Seismic Base Isolation: Necessity of Base Isolation Systems, Types of Seismic Isolation Systems, Configuration and Qualitative Behavior of Isolated Building.

Analysis of Structures with Seismic Isolation Systems: Behaviour of RC structures with Isolation Systems, Review of Building code requirements.

UNIT V

Earthquake Load Analysis on Structures: Introduction to methods of Earthquake Load Analysis (Linear Static, Linear Dynamic, Non Linear Static, Non Linear Dynamic)
Analysis of Structure by Linear Static Method (Seismic Coefficient Method)
Analysis of Structure by Linear Dynamic Method (Random Response Method)

Outcomes:

After completion of this course, students will be able to....

- Illustrate basics of Earthquake engineering viz., causes, characteristics, remedial measures
- Evaluate Single and Multi Degree of Freedom systems
- Evaluate storey shear and torsional moments for R.C. frames.
- Explain methodologies of Restoration and retrofitting of RC and masonry buildings
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Text Books:

1. Jaikrishna, Chandrasekaran and Brijesh Chandra “Elements of Earthquake Engineering”, South Asian Publishers, New Delhi, 1st edn., 1994.
2. S. K. Duggal “Earthquake Resistant Design of Structures”, Oxford University Press, New Delhi, 1st edn., 2007.
3. Anil Chopra, “Dynamics of Structures”, Prentice Hall India Publications, 3rd edn, 1995.
4. R. W. Clough and J. Penzien “Dynamics of Structures” McGraw Hill Civil Engineering Series, 3rd edn, 2003.
5. Pankaj Agarwal and Manish Shrikande, “Earthquake Resistant Design of Structures”, Prentice Hall of India Publications, 1st edn, 2006.
6. Relevant Latest Revisions of IS codes. IS 1893, IS 4326, IS 13920, IS 13827, IS 13828, IS 13935

ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

Externals: 60 Marks

L-T-P-C

Internals: 40 Marks

3-0-0-3

Objectives:

- To appreciate the purpose and role of EIA in the decision –making process
- To understand strengths & limitations of Environmental management
- To plan and carry out an environmental impact assessment study

UNIT-1 (8 hours)

Introduction to EIA: The Need for EIA, Indian Policies Requiring EIA, The EIA Cycle and Procedures, Screening, Scoping, Baseline Data, Impact Prediction, Assessment of Alternatives, Delineation of Mitigation Measure and EIA Report, Public Hearing, Decision Making, Monitoring the Clearance Conditions, Components of EIA, Roles in the EIA Process. Government of India Ministry of Environment and Forest Notification (2000), List of projects requiring Environmental clearance, Application form, Composition of Expert Committee, Ecological sensitive places, International agreements.

UNIT-II (10 hours)

Identifying the Key Issues: Key Elements of an Initial Project Description and Scoping, Project Location(s), Land Use Impacts, Consideration of Alternatives, Process selection - Construction Phase, Input Requirements, Wastes and Emissions, Air Emissions, Liquid Effluents, Solid Wastes, Risks to Environment and Human, Health, Socio-Economic Impacts, Ecological Impacts, Global Environmental Issues.

UNIT-III (12hours)

EIA Methodologies: Criteria for the selection of EIA methodology, impact identification, impact measurement, impact interpretation & Evaluation, impact communication, Methods-Adhoc methods, Checklists methods, Matrices methods, Networks methods, Overlays methods, Environmental index using factor analysis, Cost/benefit analysis, Predictive or Simulation methods. Rapid assessment of Pollution sources method, predictive models for impact assessment, Applications for RS and GIS.

UNIT-IV (8 hours)

Reviewing the EIA Report: Scope, Baseline Conditions, Site and Process alternatives, Public hearing, Construction Stage Impacts, Project Resource Requirements and Related Impacts, Prediction of Environmental Media Quality, Socio-economic Impacts, Ecological Impacts, Occupational Health Impact, Major Hazard/ Risk Assessment, Impact on Transport System, Integrated Impact Assessment.

UNIT-V (8 hours)

Review of EMP and Monitoring: Environmental Management Plan, Identification of Significant or Unacceptable Impacts Requiring Mitigation, Mitigation Plans and Relief & Rehabilitation, Stipulating the Conditions, what should be monitored? Monitoring Methods, who should monitor? Pre-Appraisal and Appraisal.

Case Studies: Preparation of EIA for developmental projects- Factors to be considered in making assessment decisions, Water Resources Project, Pharmaceutical industry, thermal plant, Nuclear fuel complex, Highway project, Sewage treatment plant, Municipal Solid waste processing plant, Tannery industry, Thermal Power Projects, Port and Harbour Projects, Mining Projects.

Outcomes:

- Identify the environmental attributes to be considered for the EIA study
- Formulate objectives of the EIA studies
- Identify the methodology to prepare rapid EIA
- Prepare EIA reports and environmental management plans

Suggested Reading:

1. Jain, R.K., Urban, L.V., Stracy, G.S., Environmental Impact Analysis, Van Nostrand Reinhold Co., New York, 1991.
2. Barthwal, R. R., Environmental Impact Assessment, New Age International Publishers, 2002
3. Rau, J.G. and Wooten, D.C., Environmental Impact Assessment, McGraw Hill Pub. Co., New York, 1996.
4. Anjaneyulu. Y and Manickam. V., Environmental Impact Assessment Methodologies, B.S. Publications, Hyderabad, 2007.
5. Wathern. P Environmental Impact Assessment- Theory and Practice, Routledge Publishers, London, 2004.

CONSTRUCTION EQUIPMENT & AUTOMATION**Externals: 60Marks****L-T-P-C****Internals: 40Marks****3-0-0-3****Objectives:**

- To introduce various construction equipment for earthwork, material handling and other miscellaneous purposes.
- To study the working of the equipment mentioned above and apply scientific principles for effectively utilizing them.

UNIT-I: (10 Lectures)

Equipment Management: Identification –Planning - Equipment Management in Projects - Maintenance Management – Replacement - Cost Control of Equipment – Depreciation Analysis, Methods of calculation of depreciation- Safety Management.

UNIT-II: (9 Lectures)

Earthwork Equipment: Fundamentals of Earth Work Operations - Earth Moving operations- Types of Earthwork Equipment - Tractors, Motor Graders, Scrapers, Front end Loaders, Earth Movers – capacity calculations.

UNIT-III: (9 Lectures)

Pumps Used in Construction: Equipment for Dredging, Trenching, Tunneling, Drilling and Blasting. Equipment for compaction - Types of pumps used in Construction - Equipment for Grouting - Pile Driving Equipment- Equipment of Erection and demolition.

UNIT-IV: (8 Lectures)

Forklifts Equipment: Forklifts and related equipment - Portable Material Bins - Conveyors - equipment used in demolition – Chain Pulley Blocks.

UNIT-V: (9 Lectures)

Screening Equipment: Crushers – Feeders - Screening Equipment - Batching and Mixing Equipment – Hauling equipment - Pouring and Pumping Equipment – Ready mixed concrete carriers

Outcomes: At the end of the course, the students will be able to

- Understand the different construction equipment for different constructions.
- Analyze the operation and maintain cost of equipment used in construction industry.

Suggested References:

1. Peurifoy, R.L., Ledbetter, W.B. and Schexnayder.C, “Construction Planning Equipment and Methods”, McGraw Hill. Singapore 2005.
2. Sharma S.C. “Construction Equipment and Management”, Khanna Publishers, Delhi, 2008.
3. Deodhar, S.V. “Construction Equipment and Job Planning”, Khanna Publishers Delhi, 2008.
4. Mahesh VarmaDr., “Construction Equipment and its planning and application”, Metropolitan Book Company, New Delhi, 2003

PAVEMENT CONSTRUCTION AND MANAGEMENT

Externals: 60Marks

L-T-P-C

Internals: 40Marks

3-0-0-3

Objectives:

- To understand the construction techniques of flexible and rigid pavements
- To evaluate the pavement functionally and structurally
- To understand the pavement management system concepts

UNIT I:

Flexible Pavement Construction: Earthwork, compaction and construction of embankments, specifications of materials, construction methods and field control checks for various types of flexible pavement materials in sub-base, base, binder and surface course layers and their choice; Cement Concrete Pavement Layers:

UNIT II:

Specifications and method of cement concrete pavement construction; Construction of interlocking block pavements, Quality control tests; Construction of various types of joints; Soil Stabilized Pavement Layers: Principles of gradation/proportioning of soil-aggregate mixes and compaction; Design factors, mix design, construction control and quality control checks for mechanical, soil-cement, soil-bitumen and soil-lime stabilization methods. Use of additives, Numerical problems on mix design and applications;

UNIT III:

Pavement Evaluation - Pavement Distress - Functional and structural condition of pavements, Pavement distress survey, Functional condition evaluation of pavements- Roughness, Skid Resistance.

UNIT IV:

Structural evaluation of pavements - nondestructive testing, Benkelman beam and Falling Weight Deflectometer, Pavement strengthening based on deflection as per IRC, Maintenance and rehabilitation techniques;

UNIT V:

Pavement Management Systems - Pavement Management Systems- Components, structure, data requirements, Project level and Network level needs, Pavement performance prediction – concepts, modelling techniques– AASTHO, CRRI and HDM models, Budget forecasting for maintenance and rehabilitation, Ranking and optimization methodologies, life cycle costing.

Outcome:

Student will be able to understand different techniques for pavement evaluation and maintenance procedure of pavements

Suggested Reading:

1. Khanna, S. K. and Justo, C. E. G (1994), "Highway Engineering", Nemchand & Bros, New Delhi. India.
2. McShane, W.R., Roess, R.P. and Prassas, E.S., Traffic Engineering. Prentice Hall. Englewood Cliffs, 1997.
3. Highway Capacity Manual, Transportation Research Board, National Research Council. Washington, D.C., 2000.
4. L.R. Kadiyali, *Principles and Practice of Highway Engineering*, Khanna Publications, New Delhi, 2000
5. Haas and Hudson W. R., "Pavement Management Systems" McGraw Hill publications 2004
6. Y.H. Huang, *Pavement Analysis and Design*, Prentice Hall, Englewood Cliffs, New Jersey, 2004

PRESTRESSED CONCRETE

Externals: 60Marks

Internals: 40Marks

L-T-P-C

3-0-0-3

Objectives:

- Elements of prestressed concrete and systems of prestressing
- Losses and analysis of prestressed concrete
- Analysis and design of section in flexure and shear
- Analysis and transfer of prestress

UNIT –I: (14hours)

Elements of Prestressed concrete: Historical development, concepts of pre-stressing, Terminology, basic definitions, Advantages, Applications of prestressed concrete, Pre-tensioning and Post-tensioning.

Prestressing Systems: Hoyer, Freyssinet, Magnel blaton and Lee-Mc call system, material properties, Need for High strength steel and High strength concrete.

UNIT –II: (14hours)

Losses of Prestress: Loss of prestress in pre-tensioned and post-tensioned members due to various causes like elastic shortening of concrete, shrinkage of concrete, creep of concrete, relaxation of steel, slip in anchorage, bending of member and frictional loss.

Analysis of Sections: Stress concept, Strength concept, Load balancing concept, Effect of loading on the tensile stresses in tendons, Effect of tendon profile on deflections, Factors influencing deflections, Calculation of Short term and long term deflections.

UNIT –III: (14hours)

Design of Section for Flexure: Allowable stresses, Elastic design of simple beams having rectangular and I-section for flexure, kern lines, Cable profile and cable layout.

Design of Section for Shear: Shear and Principal Stresses, improving shear resistance by different prestressing techniques, horizontal, sloping and vertical prestressing, Analysis of rectangular and I-beam, Design of shear reinforcement, Indian code provisions.

UNIT –IV: (10hours)

Transfer of Prestress: Transmission of prestressing force by bond, Transmission length, Flexural bond stresses, IS code provisions, Anchorage zone stresses in post tensioned members, stress distribution in End block. Methods of transfer: Analysis by approximate, Guyon and Magnel methods, Anchorage zone reinforcement.

UNIT –V: (8hours)

Analysis of continuous beams: Elastic theory- Linear transformation and Concordant tendons Deflections of pre-stressed concrete beams: Importance of control of deflections- factors influencing deflections-short term deflections of un-cracked member – prediction of long term deflections

Outcomes:

- Appraise the concept of prestressing in concrete
- Analyze prestressed concrete sections
- Design prestressed concrete members in flexure and shear

Suggested Reading:

1. Prestressed Concrete- N. Krishna Raju
2. Prestressed Concrete- S. Ramamrutham
3. Prestressed Concrete- P. Dayaratnam
4. Prestressed Concrete- T.Y.Lin
5. I.N. Krishna Raju, “Prestressed concrete”, Tata McGraw Hill Company, New Delhi, 5th edn., 2012
6. S. K. Mallic and A. P. Gupta, “Prestressed concrete”, Oxford and IBH publishing Co. Pvt. Ltd., 4th edn., 1987

CIVIL ENGINEERING- SOCEITAL AND GLOBAL IMPACT

Externals: 60Marks

L-T-P-C

Internals: 40Marks

2-0-0-2

Objectives:

The course is designed to provide a better understanding of

- The impact which Civil Engineering has on the Society at large and on the global arena.
- Civil Engineering projects have an impact on the Infrastructure, Energy consumption and generation, Sustainability of the Environment, Aesthetics of the environment, Employment creation, Contribution to the GDP, and on a more perceptible level, the Quality of Life.
- It is important for the civil engineers to realise the impact which this field has and take appropriate precautions to ensure that the impact is not adverse but beneficial.

Module 1: Introduction to Course and Overview; Understanding the past to look into the future: Preindustrial revolution days, Agricultural revolution, first and second industrial revolutions, IT revolution; Recent major Civil Engineering breakthroughs and innovations; Present day world and future projections, Ecosystems in Society and in Nature; the steady erosion in Sustainability; Global warming, its impact and possible causes; Evaluating future requirements for various resources; GIS and applications for monitoring systems; Human Development Index and Ecological Footprint of India Vs other countries and analysis;

Module 2: Understanding the importance of Civil Engineering in shaping and impacting the world; The ancient and modern Marvels and Wonders in the field of Civil Engineering; Future Vision for Civil Engineering

Module 3: Infrastructure - Habitats, Megacities, Smart Cities, futuristic visions; Transportation (Roads, Railways & Metros, Airports, Seaports, River ways, Sea canals, Tunnels (below ground, under water); Futuristic systems (ex, Hyper Loop)); Energy generation (Hydro, Solar (Photovoltaic, Solar Chimney), Wind, Wave, Tidal, Geothermal, Thermal energy); Water provisioning; Telecommunication needs (towers, above-ground and underground cabling); Awareness of various Codes & Standards governing Infrastructure development; Innovations and methodologies for ensuring Sustainability;

Module 4: Environment-Traditional & futuristic methods; Solid waste management, Water purification, Wastewater treatment & Recycling, Hazardous waste treatment; Flood control (Dams, Canals, River interlinking), Multi-purpose water projects, Atmospheric pollution; Global warming phenomena and Pollution Mitigation measures, Stationarity and non-stationarity; Environmental Metrics & Monitoring; Other Sustainability measures; Innovations and methodologies for ensuring Sustainability.

Module 5: Built environment–Facilities management, Climate control; Energy efficient built

environments and LEED ratings, Recycling, Temperature/ Sound control in built environment, Security systems; Intelligent/ Smart Buildings; Aesthetics of built environment, Role of Urban Arts Commissions; Conservation, Repairs & Rehabilitation of Structures & Heritage structures; Innovations and methodologies for ensuring Sustainability

Module 6: Civil Engineering Projects – Environmental Impact Analysis procedures; Waste (materials, manpower, equipment) avoidance/ Efficiency increase; Advanced construction techniques for better sustainability; Techniques for reduction of Green House Gas emissions in various aspects of Civil Engineering Projects; New Project Management paradigms & Systems (Ex. Lean Construction), contribution of Civil Engineering to GDP, Contribution to employment (projects, facilities management), Quality of products, Health & Safety aspects for stakeholders; Innovations and methodologies for ensuring Sustainability during Project development;

Course outcomes:

- The impact which Civil Engineering projects have on the Society at large and on the global arena and using resources efficiently and effectively.
- The extent of Infrastructure, its requirements for energy and how they are met: past, present and future
- The Sustainability of the Environment, including its Aesthetics,
- The potentials of Civil Engineering for Employment creation and its Contribution to the GDP
- The Built Environment and factors impacting the Quality of Life
- The precautions to be taken to ensure that the above-mentioned impacts are not adverse but beneficial.
- Applying professional and responsible judgement and take a leadership role;

Text/Reference Books:

1. Brito, Ciampi, Vasconcelos, Amarol, Barros (2013) Engineering impacting Social, Economical and Working Environment, 120th ASCE Annual Conference and Exposition
2. NAE Grand Challenges for Engineering (2006), Engineering for the Developing World, The Bridge, Vol 34, No.2, Summer 2004.
3. Allen M. (2008) Cleansing the city. Ohio University Press. Athens Ohio.
4. Ashley R., Stovin V., Moore S., Hurley L., Lewis L., Saul A. (2010). London Tideway Tunnels Programme – Thames Tunnel Project Needs Report – Potential source control and SUDS applications: Land use and retrofit options
5. <http://www.thamestunnelconsultation.co.uk/consultation-documents.aspx>
6. Ashley R M., Nowell R., Gersonius B., Walker L. (2011). Surface Water Management and Urban Green Infrastructure. Review of Current Knowledge. Foundation for Water Research FR/R0014
7. Barry M. (2003) Corporate social responsibility – unworkable paradox or sustainable paradigm? Proc ICE Engineering Sustainability 156. Sept Issue ES3 paper 13550. p 129-130
8. Blackmore J M., Plant R A J. (2008). Risk and resilience to enhance sustainability with application to urban water systems. J. Water Resources Planning and Management. ASCE. Vol. 134, No. 3, May.
9. Bogle D. (2010) UK's engineering Council guidance on sustainability. Proc ICE Engineering

Sustainability 163. June Issue ES2 p61-63

10. Brown R R., Ashley R M., Farrelly M. (2011). Political and Professional Agency Entrapment: An Agenda for Urban Water Research. Water Resources Management. Vol. 23, No.4. European Water Resources Association (EWRA) ISSN 0920-4741.

11. Brugnach M., Dewulf A., Pahl-Wostl C., Taillieu T. (2008) Toward a relational concept of uncertainty: about knowing too little, knowing too differently and accepting not to know. Ecology and Society 13 (2): 30

12. Butler D., Davies J. (2011). Urban Drainage. Spon. 3rd Ed.

13. Cavill S., Sohail M. (2003) Accountability in the provision of urban services. Proc. ICE. Municipal Engineer 156. Issue ME4 paper 13445, p235-244.

14. Centre for Water Sensitive Cities (2012) Blueprint for a water sensitive city. Monash University.

15. Charles J A. (2009) Robert Rawlinson and the UK public health revolution. Proc ICE Eng History and Heritage. 162 Nov. Issue EH4. p 199-206